

VIETNAM-KOREA UNIVERSITY OF INFORMATION
AND COMMUNICATION TECHNOLOGY
FACULTY OF COMPUTER SCIENCE



GRADUATION THESIS

**BUILDING A CROSS-PLATFORM INTELLIGENT TOURISM
SYSTEM USING AI**

Student: Nguyen Thanh Cong **Class:** 21KIT
Industry: Information Technology
Major: Software Engineering
Supervisor: MSc. Nguyen Ngoc Huyen Tran
Eng. Tu Tan Hoang Son

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SUPERVIROR’S COMMENT

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Nguyen Thanh Cong

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ABBREVIATIONS

ABBREVIATIONS	MEANING
CV	Curriculum Vitae
AI	Artificial Intelligence
API	Application Programming Interface
SQL	Structured Query Language
I/O	Input/output

INTRODUCTION

1. PROBLEM STATEMENT AND RATIONALE

The tourism industry is experiencing a digital revolution, with users increasingly expecting personalized and seamless experiences across platforms. However, the overwhelming amount of information and service options available online often leads to confusion and decision fatigue for travelers. Service providers also face challenges in reaching the right customers and managing their offerings efficiently. Therefore, there is a pressing need for an intelligent system that can bridge the gap between user expectations and service delivery. By leveraging artificial intelligence specifically recommendation systems and chatbots this project aims to create a smart ecosystem that not only streamlines the user journey but also empowers providers to deliver better, more targeted services. This approach promises to enhance user satisfaction, optimize resource allocation, and drive innovation in the tourism sector.

2. PROJECT OBJECTIVES

The main objective of this project is to design and implement the Tripfinity ecosystem-a comprehensive, cross-platform solution for intelligent tourism management. The system consists of three tightly integrated components:

A mobile application (built with Flutter) for end-users, providing features such as service discovery, personalized recommendations, seamless booking, and real-time AI chatbot support for travel planning and assistance.

A web-based management portal (developed with React) for service providers, enabling them to efficiently manage their listings, handle bookings, respond to customer feedback, and analyze business performance.

A robust backend system (using Java Spring Boot and Python) that handles business logic, data storage, security, and integrates advanced AI modules for recommendations and conversational support. This backend exposes APIs to both the mobile and web frontends, ensuring smooth and secure data flow.

The overarching goal is to automate and personalize the tourism experience, making it more efficient, user-friendly, and scalable for future growth.

3. SCOPE AND OBJECT OF STUDY

Object of Study:

This project centers on the design and implementation of AI-driven recommendation systems, intelligent chatbots, and comprehensive workflows for

tourism service management including tours, hotels, attractions, and restaurants. It also examines the challenges of large-scale data processing, user experience optimization, and effective communication between customers and service providers.

Scope:

The mobile application is developed using Flutter, delivering a seamless and consistent user experience on both Android and iOS devices.

The supplier management website is built with React, focusing on intuitive interfaces, scalability, and efficient service operations for providers.

The backend system utilizes Java Spring Boot for business logic, security, and data management, while Python is employed for developing machine learning models and natural language processing modules for the chatbot.

The overall system is enhanced by integrating advanced technologies such as Firebase (for authentication and notifications), ZaloPay (for payment processing), Cloudinary (for media management), and Google Maps (for geolocation services), resulting in a robust, modern, and feature-rich solution.

4. RESEARCH METHODOLOGY

The project employs a combination of theoretical research and practical experimentation:

In-depth study of deep learning techniques, with a focus on Two-Tower Neural Network architectures for personalized recommendations and Retrieval-Augmented Generation (RAG) for advanced chatbot capabilities.

Requirement analysis and system design, adopting a modular, microservices-based architecture to ensure flexibility and scalability.

Iterative development and testing of each system component, including backend APIs, AI models, and user interfaces for both mobile and web platforms.

Application of big data processing methods to handle large-scale user and service data, continuous optimization of AI models for accuracy and performance, and deployment in real-world scenarios.

Ongoing evaluation using key metrics such as recommendation accuracy, system latency, user engagement, and feedback to ensure the solution meets practical needs and delivers measurable value.

5. PRACTICAL SIGNIFICANCE

This project delivers significant practical benefits for both travelers and service providers:

For users, it addresses the problem of information overload by offering intelligent, context-aware recommendations and instant support, making it easier to discover, compare, and book the most suitable tourism services.

For providers, it facilitates digital transformation by streamlining service management, improving customer engagement, and enabling data-driven decision-making.

On a broader scale, the project demonstrates the real-world application of AI in the tourism industry in Vietnam, setting a foundation for future advancements in smart, personalized, and automated tourism solutions. It contributes to the modernization of the sector, enhances competitiveness, and supports the sustainable growth of digital tourism.

CHAPTER 1. THEORETICAL BASIS AND TECHNOLOGY

1.1. OVERVIEW OF DEVELOPMENT TECHNOLOGIES

1.1.1. Mobile and Web Technologies (Flutter & ReactJS)

Flutter (Mobile)

Flutter is an open-source UI development toolkit developed by Google, enabling the creation of cross-platform mobile applications (Android, iOS) with a single codebase. The decision to use Flutter for the Tripfinity project is based on the following reasons:

Cross-platform capability: Flutter allows development for multiple operating systems without rewriting code, saving both time and resources.

High performance: Flutter uses the Dart language, which compiles directly to native machine code, delivering smooth performance suitable for dynamic, interactive applications like Tripfinity.

Beautiful and flexible UI: Flutter provides a rich set of customizable widgets, enabling the creation of modern, user-friendly interfaces.

Large community and extensive documentation: A strong developer community makes it easy to find solutions and stay updated with the latest technologies.

In Tripfinity, Flutter is used to develop the mobile application for travelers, providing features such as search, service booking, personalized recommendations, and AI chatbot interaction.

ReactJS + TypeScript (Web Supplier)

ReactJS is a popular JavaScript library developed by Facebook, combined with TypeScript (a superset of JavaScript that supports static type checking) to build modern web interfaces. The reasons for choosing ReactJS + TypeScript for Tripfinity include:

Component-based architecture: ReactJS enables the interface to be divided into independent, reusable, and maintainable components.

Type safety: TypeScript helps detect errors during development, minimizing risks when implementing complex features.

Scalability: ReactJS is well-suited for management dashboards that require frequent data updates and future feature expansion.

Robust ecosystem: Many supporting libraries are available, making it easy to integrate with modern technologies such as Redux, Material UI, etc.

The Tripfinity Web Supplier is built with ReactJS + TypeScript, serving partners (travel service providers) in managing products, orders, monitoring statistics, and interacting with the AI system.

1.1.2. Backend Technologies (Spring Boot & MySQL)

Spring Boot:

Spring Boot is a powerful Java framework that supports backend development following the Microservices architecture. The use of Spring Boot in Tripfinity brings several advantages:

Microservices architecture: Allows the system to be divided into independent services, making it easier to scale, maintain, and deploy.

RESTful API: Spring Boot supports the development of standard REST APIs, facilitating efficient communication between frontend and backend, and easy integration with other systems.

Security: Integration with Spring Security helps protect the system, authenticate users via JWT, and control access by roles.

Easy AI service integration: The backend can connect to AI services (Python) via REST API or gRPC, ensuring flexibility and scalability.

In Tripfinity, Spring Boot is responsible for handling core business logic such as user management, service booking, authentication, and connecting to the AI system for recommendations and chatbot responses.

MySQL

MySQL is a widely used relational database management system, chosen for Tripfinity for the following reasons:

Ensuring consistency (ACID): MySQL supports transactions that guarantee data integrity, which is crucial for service booking and payment operations.

High performance and scalability: MySQL can handle large volumes of data and is easily scalable as the system grows.

Popularity and easy integration: MySQL is well-supported by frameworks like Spring Boot and can be deployed on various platforms.

In Tripfinity, the MySQL database stores user information, services, orders, and interaction history, serving both the backend and AI models.

1.1.3. Third-party Integrations (Google Maps, ZaloPay, Cloudinary)

To enhance user experience and provide essential features for a modern travel platform, Tripfinity integrates several third-party services as follows:

Google Maps:

The Google Maps API is integrated to display interactive maps and provide address search functionality, allowing users to view the locations of travel-related services on the map. Users can enter an address to see its position and easily look up related service locations (such as hotels, restaurants, and attractions) directly within the application. This integration improves user experience by supporting convenient and intuitive location lookup.

ZaloPay:

ZaloPay is a popular electronic payment gateway in Vietnam. By integrating ZaloPay, Tripfinity enables users to make secure and convenient online payments when booking tours, hotels, and other services. This integration ensures fast transaction processing, increases reliability, and meets the payment habits of the local market.

Cloudinary:

Cloudinary is a cloud-based media management service used for storing, optimizing, and delivering images and videos. In Tripfinity, Cloudinary is utilized to manage user-uploaded photos, service images, and promotional media content. This integration ensures fast loading speeds, automatic image optimization, and stable content delivery across various platforms and devices.

Groq API:

The Groq API is integrated to provide powerful Natural Language Processing (NLP) capabilities for chatbot features and intelligent support in Tripfinity. Through the Groq API, the system can understand and respond to user queries naturally, assist with information search, suggest services, and enhance interactive experiences. This integration accelerates processing speed, ensures accuracy, and expands personalization capabilities for users, enabling more dynamic conversations, seamless user engagement, and scalable AI-driven solutions that adapt to evolving customer needs and preferences.

1.2. ARTIFICIAL INTELLIGENCE FOUNDATIONS

1.2.1. Recommender Systems

The recommender system is a crucial component in digital tourism platforms, personalizing user experiences by suggesting services and itineraries that match individual preferences and behaviors. In Tripfinity, the recommender system is built on deep learning models, specifically the Two-Tower neural network architecture.

Two-Tower Neural Network Architecture:

User Tower: A neural network that represents user features based on booking history, interaction behavior, and personal attributes.

Item Tower: A neural network that represents features of each service (tour, hotel, restaurant, etc.) based on attributes, descriptions, etc.

Matching mechanism: The feature vectors from the User Tower and Item Tower are compared (typically using cosine similarity or dot product) to calculate relevance scores, thereby generating an optimal recommendation list for each user.

Loss function: Loss functions such as binary cross-entropy or triplet loss are used to optimize the model.

Training pipeline: Data is collected, processed, and trained on the Google Colab platform. The trained model is saved in .keras format for integration into the production system.

1.2.2. Natural Language Processing (NLP) and Chatbots

Large Language Model (LLM):

LLMs are AI models capable of understanding and generating natural language at a high level, applied in chatbots for consultation, answering questions, and providing 24/7 customer support. In Tripfinity, the chatbot is built on state-of-the-art LLMs, enhancing user experience and reducing the workload for customer service teams.

Retrieval-Augmented Generation (RAG) Technique:

RAG is a method that combines the generative capabilities of LLMs with retrieval from proprietary knowledge sources (database, FAQ, internal documents). When a user asks a question, the system will:

Retrieve relevant information from the database or documents.

Combine it with the power of the LLM to generate natural, accurate, and up-to-date responses.

Applying RAG enables Tripfinity's chatbot to provide not only flexible but also precise answers, tailored to the tourism business context.

Groq API Integration:

The Groq API is utilized to power the chatbot's advanced NLP and conversational abilities. By leveraging Groq's high-performance language models, Tripfinity ensures faster, more accurate, and context-aware responses for users.

1.3. DEVELOPMENT SUPPORT TOOLS

1.3.1. Development Environments (IDE):

Android Studio: Used for developing and debugging Flutter applications.

IntelliJ IDEA: Used for backend development with Spring Boot.

Visual Studio Code: Used for frontend development with ReactJS and for quick editing of configuration files and AI Python scripts.

1.3.2. Testing Tools:

Postman: Used for backend API testing, business flow validation, and security checks.

Google Colab: Used for training and experimenting with AI models, and processing large datasets with free GPU resources.

1.3.3. Source Code Management:

Git: Used for version control and team collaboration.

1.3.4. Design and Prototyping:

Figma: Used for designing user interfaces, creating prototypes, and collaborating on UI/UX design with the team.

1.4. SUMMARY

Chapter 1 presents the theoretical foundation and technological choices for Tripfinity. It highlights the use of Flutter for cross-platform mobile apps, ReactJS with TypeScript for the web interface, and a Spring Boot microservices backend with MySQL. Integrations like Google Maps, ZaloPay, Clouinary, and Groq API enhance functionality and AI features. The chapter also covers AI applications, including deep learning-based recommender systems (Two-Tower architecture) and advanced chatbots using large language models and RAG, along with essential development tools for efficient implementation and collaboration.

CHAPTER 2. SYSTEM ANALYSIS AND DESIGN

2.1. SYSTEM REQUIREMENTS

2.1.1. Non-Functional Requirements

The Tripfinity system must meet the following requirements:

Minimize manual operations by automating core processes such as booking confirmation, payment handling, and notification delivery.

Support timely updates through real-time or near real-time data synchronization across web and mobile platforms.

Ensure that most user requests are processed and updated within 2–3 seconds to provide a smooth user experience.

Display search results within 3–5 seconds to allow users to quickly find suitable travel options.

Store user and booking information for at least 1 year, with a basic backup mechanism to ensure data safety and recovery.

Provide a graphical user interface that is user-friendly, simple, and visually appealing to enhance overall usability.

Protect user information by applying standard security measures, including secure authentication, encrypted data storage, and HTTPS communication.

2.1.2. Functional Requirements

2.1.2.1. Authentication & Account Management

Registration: Create account with email/password or OAuth (Google, GitHub) with profile information

Login: Authenticate using email/password or OAuth providers with JWT token generation

Profile Management: View and update user information, preferences, and provider business details

Logout: Securely terminate user session and remove authentication tokens

2.1.2.2. Service Management (Tours, Hotels, Attractions, Restaurants)

For Tourists:

Search & Discovery: Universal search with filters (location, price, amenities), map-based and GPS nearby search

View Details: Display service information including description, pricing, photos, location, reviews, and availability

Booking: Select options, calculate costs, and create bookings with payment processing

Payment: Process payments via ZaloPay or pay-at-counter with automatic status updates

Reviews: Rate services on multiple criteria, write reviews, upload photos, and like helpful reviews

For Service Providers:

Service Management: Create, edit, delete, and manage service listings with pricing and availability

Booking Management: View, confirm, or reject reservations and track booking statistics

Customer Engagement: Respond to reviews and monitor performance analytics

2.1.2.3. Trip Planning & Itinerary

Trip Creation: Create travel plans with dates, cover images, and trip types

Itinerary Management: Add services to daily schedules, drag-and-drop rearrange, set times and notes

2.1.2.4. Communication Features

AI Chatbot: 24/7 RAG-based assistant for trip planning with Vietnamese/English support and context memory

Provider Messaging: Real-time chat with service providers for inquiries and support

2.1.2.5. Additional Features

Favorites: Bookmark services and organize by category for quick access

Notifications: Push and in-app notifications for bookings, payments, and promotions

Gamification: Earn points and unlock badges through bookings, reviews, and activities

Blog: Browse and create travel content, guides, and promotional posts

2.1.3. Actors List

The TRIPFINITY system has two primary actors: Tourist and Service Provider

2.1.3.1. Tourist (User)

Individuals using the system to discover destinations, book tourism services, and plan travel experiences.

Roles:

Register and log in to the system using email/password or OAuth (Google)

Search and filter tourism services (tours, hotels, attractions, restaurants)

View detailed service information including photos, maps, reviews, and ratings

Book services and process payments via ZaloPay or pay-at-counter

Create and manage personal travel itineraries

Write reviews and rate services with photo uploads

Save favorite services and manage wishlists

Chat with AI chatbot for trip planning assistance

Communicate with service providers via real-time messaging

Earn points and unlock achievement badges

View notifications about bookings, payments, and promotions

Read and create travel blog posts

Goals:

Find suitable tourism services matching personal preferences and budget

Plan organized and memorable trips with detailed itineraries

Book reliable and high-quality services with secure payment

Share travel experiences through reviews and blog posts

Receive personalized AI-powered service recommendations

2.1.3.2. Service Provider (Supplier)

Organizations or individuals offering tourism services on the platform (tour operators, hotels, attraction managers, restaurants).

Roles:

Register and manage provider accounts with business information

Create, edit, and manage service listings (tours, hotels, attractions, restaurants)

Set pricing, availability, schedules, and booking policies

View and manage customer bookings

Confirm or reject reservation requests

Respond to customer reviews and feedback

Chat with customers via messaging system

Create and publish blog content to promote services

Monitor performance statistics and revenue reports

Receive notifications about new bookings and customer inquiries

Goals:

Reach more potential customers through the platform

Efficiently manage bookings and reservations across all services

Build strong reputation through positive customer reviews

Increase service visibility and booking conversion rates

Monitor business performance and revenue analytics

2.2. USE CASE DIAGRAM

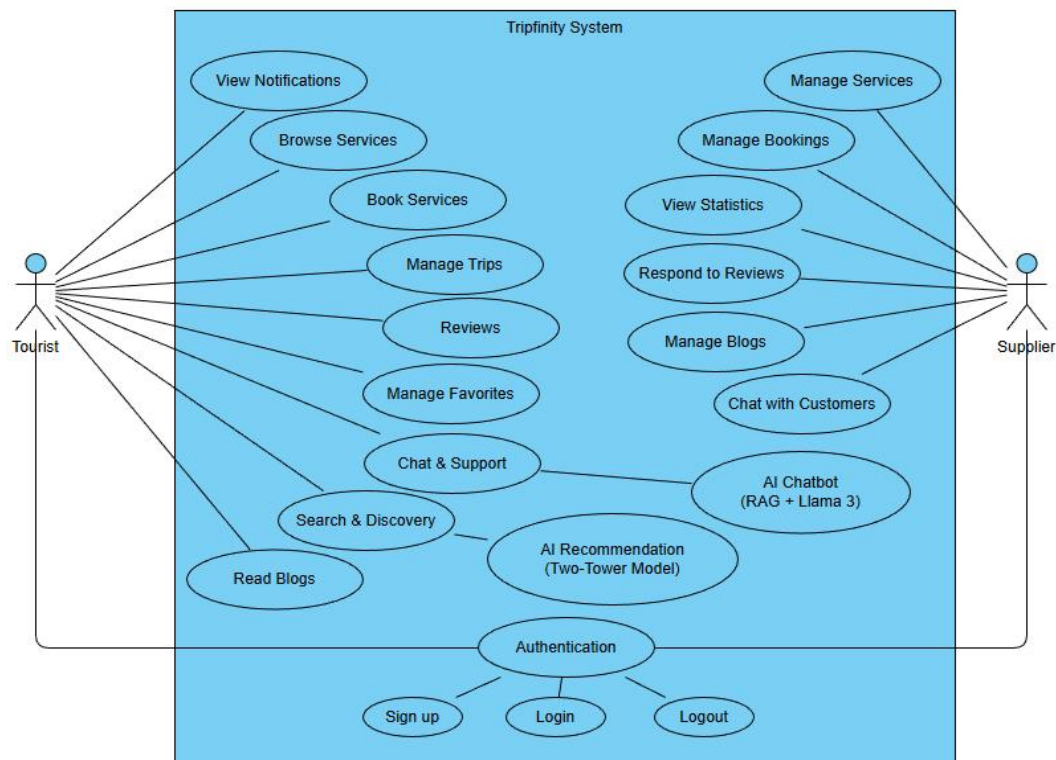


Figure 2.1. Use case diagram

2.2.1. Tourist

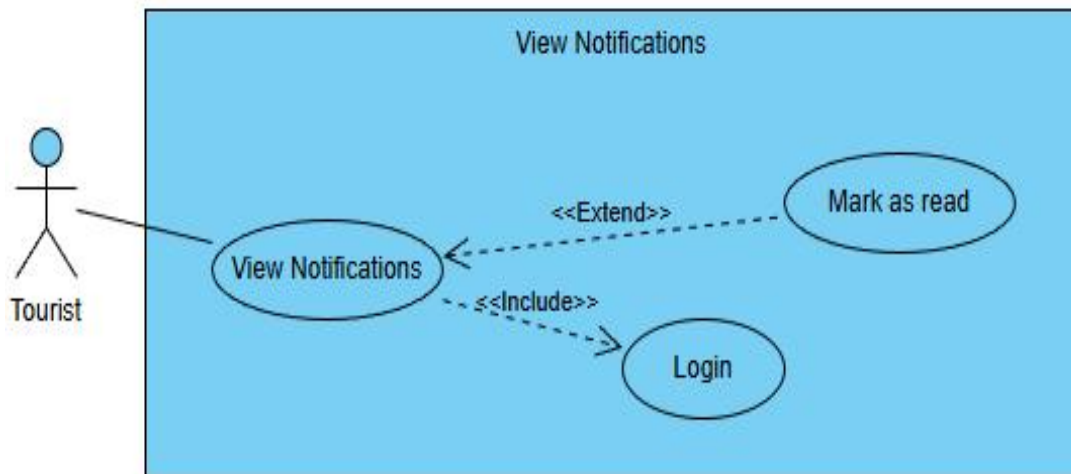


Figure 2.2. Use case View Notifications

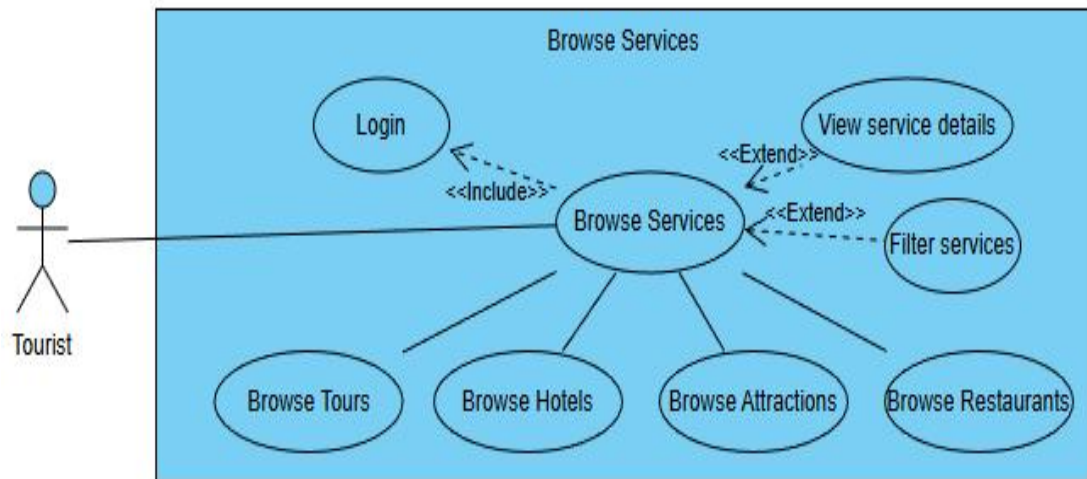


Figure 2.3. Use case Browse Services

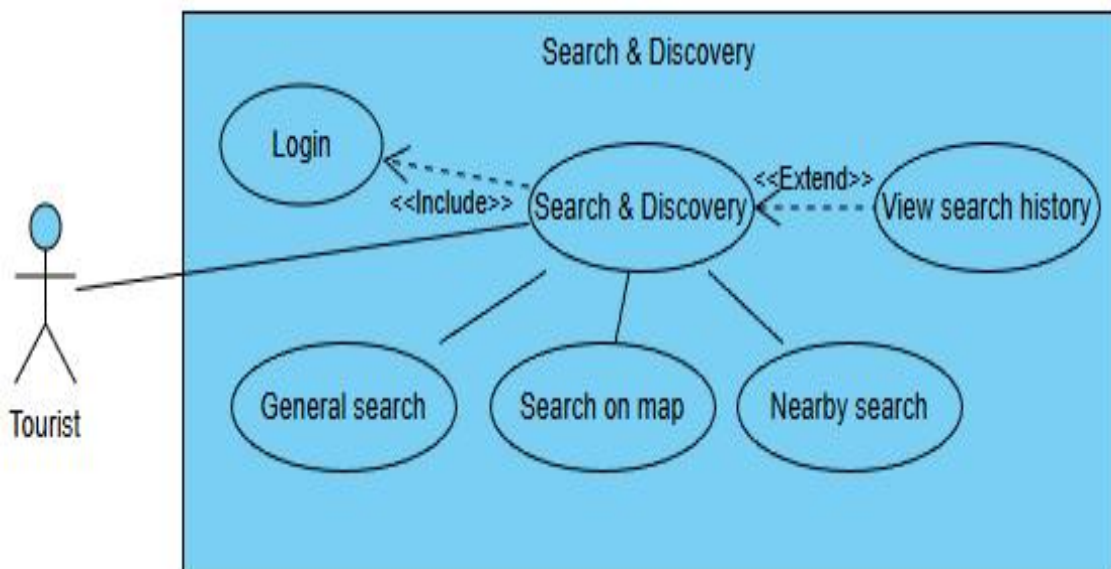


Figure 2.4. Use case Search & Discovery

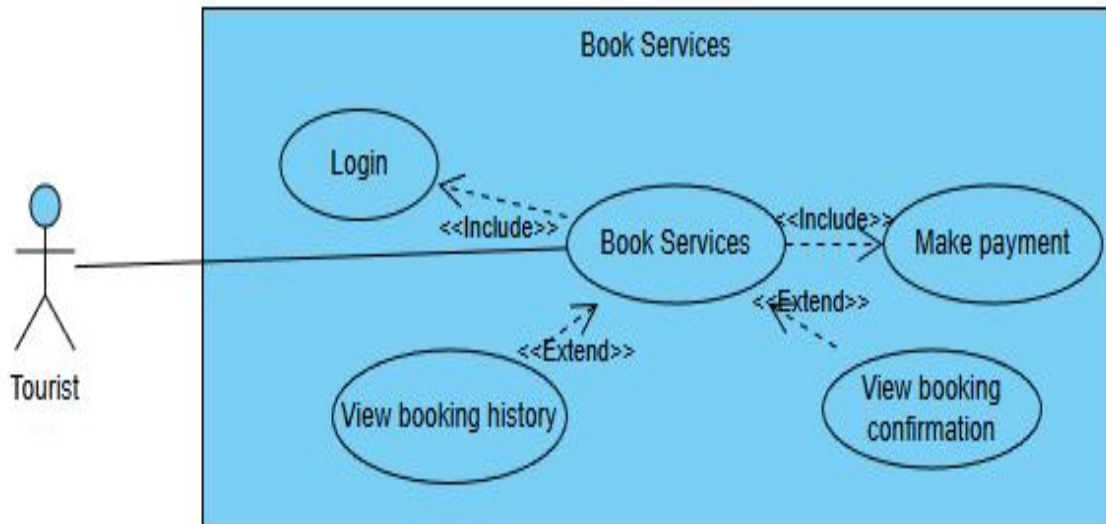


Figure 2.5. Use case Book Services

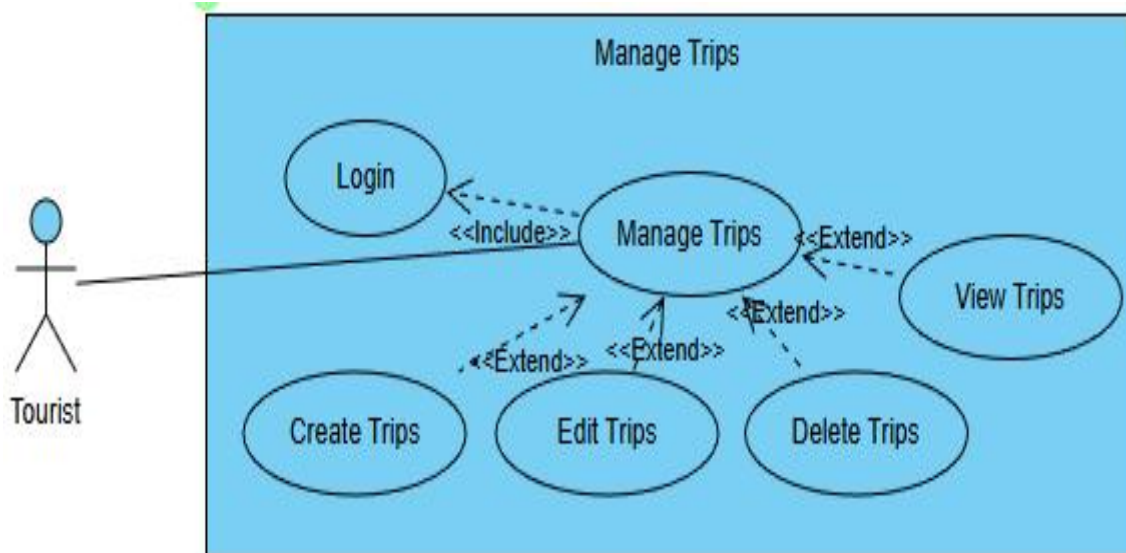


Figure 2.6. Use case Manage Trips

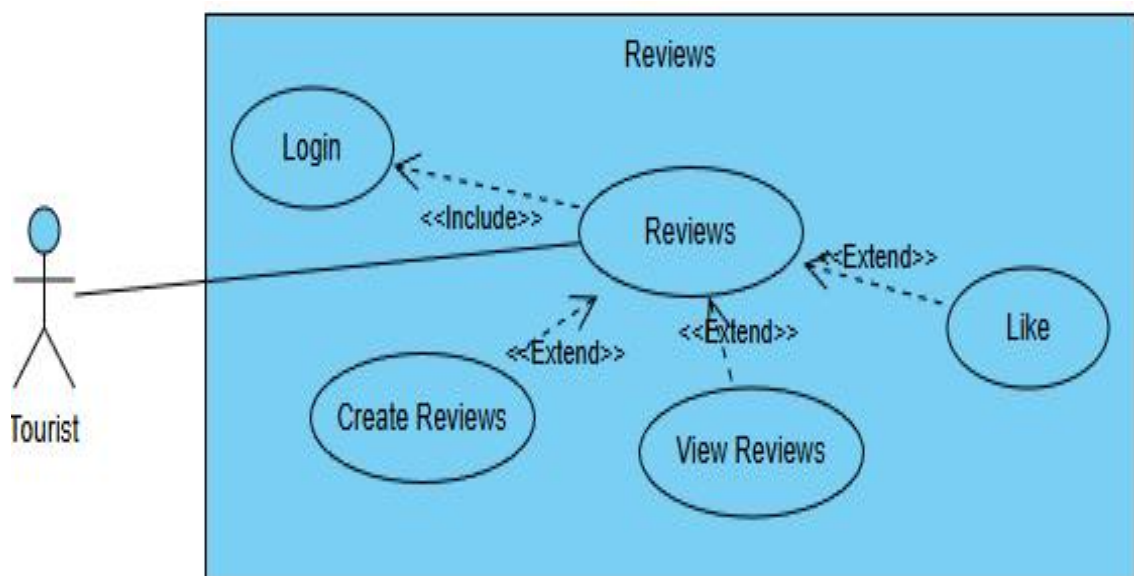


Figure 2.7. Use case Reviews

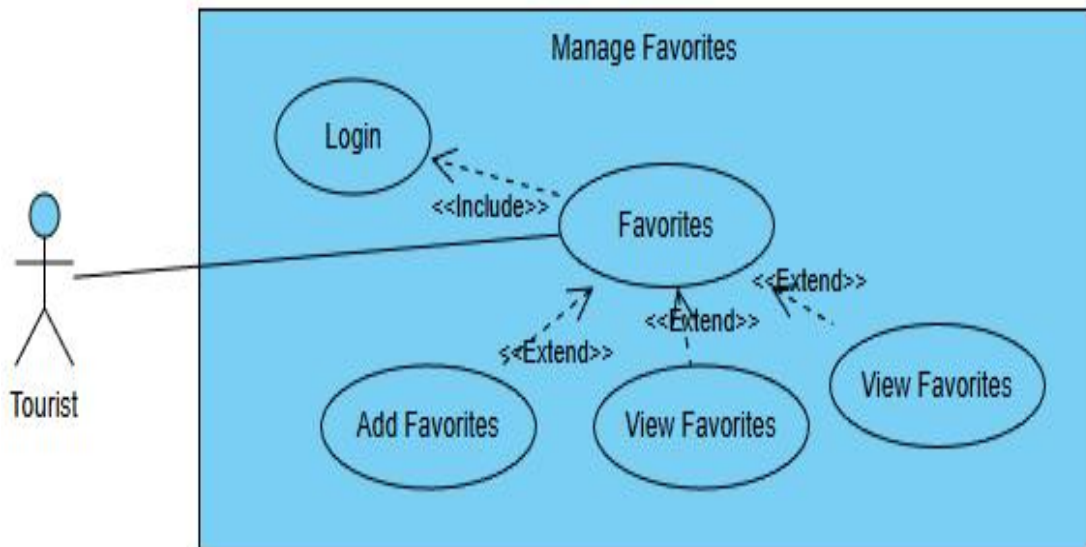


Figure 2.8. Use case Manage Favorites

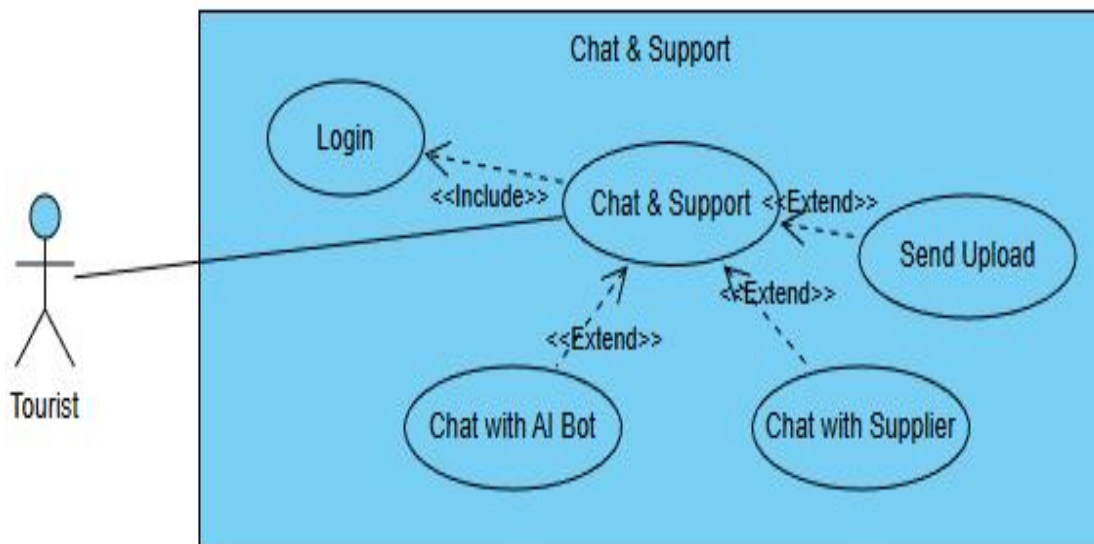


Figure 2.9. Use case Chat & Support

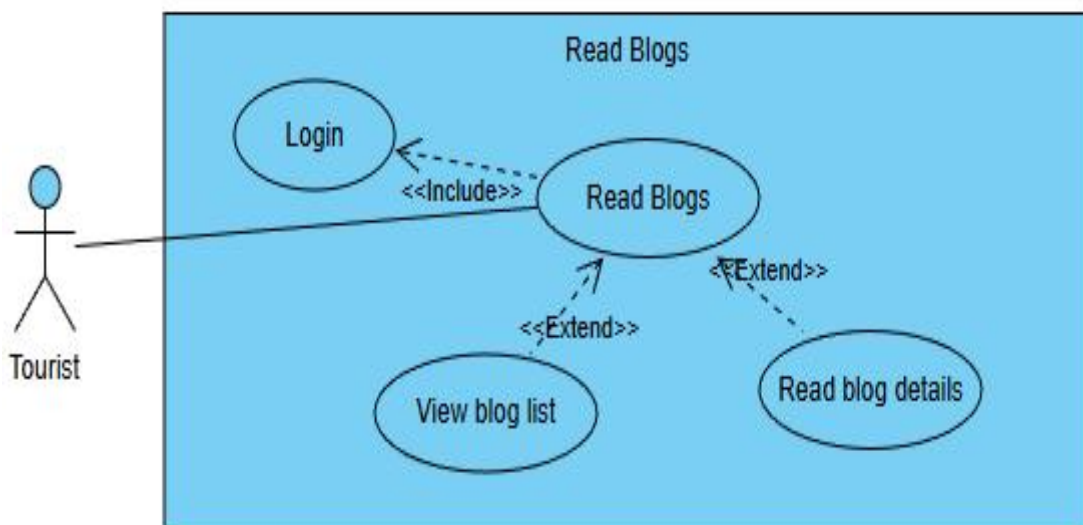


Figure 2.10. Use case Read Blogs

2.2.2. Supplier

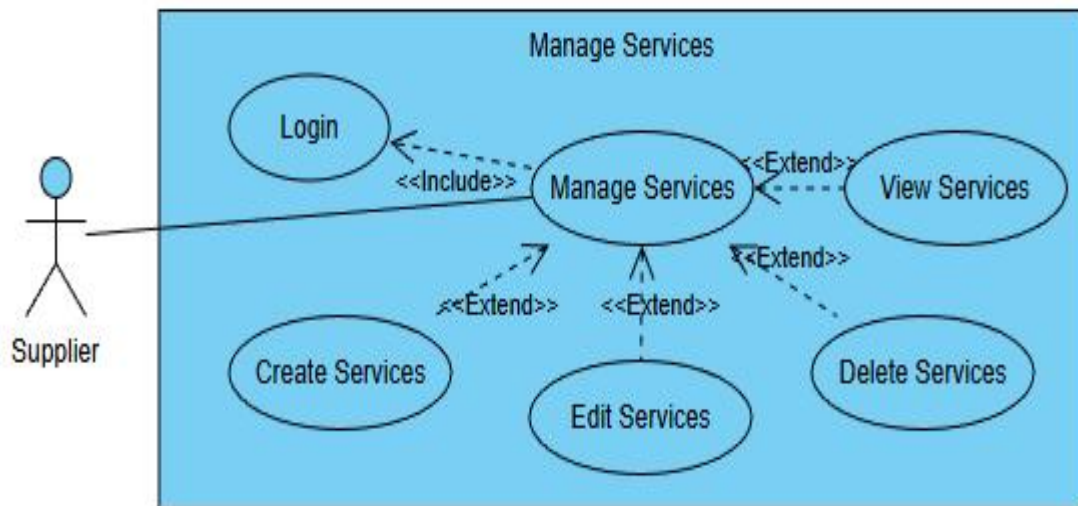


Figure 2.11. Use case Manage Services

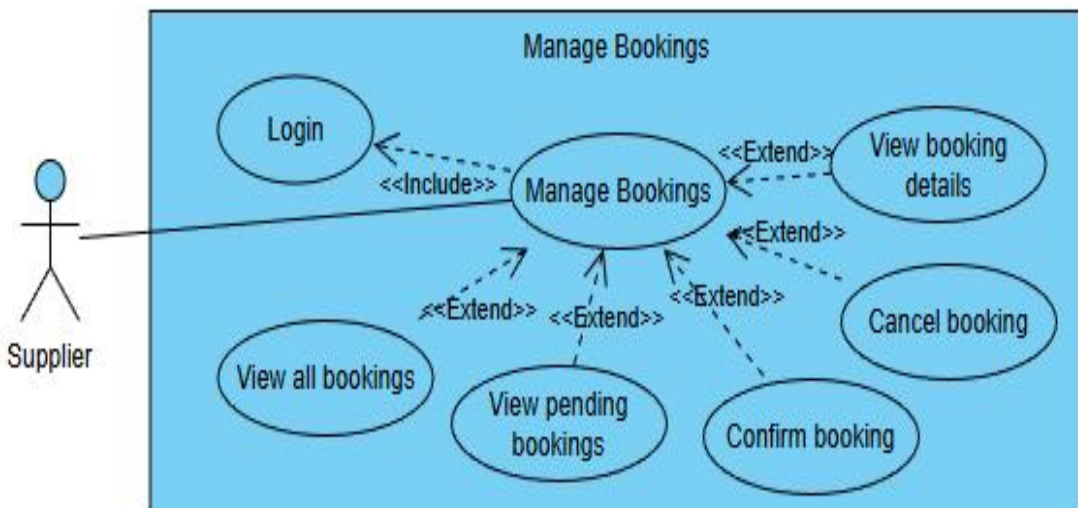


Figure 2.12. Use case Manage Bookings

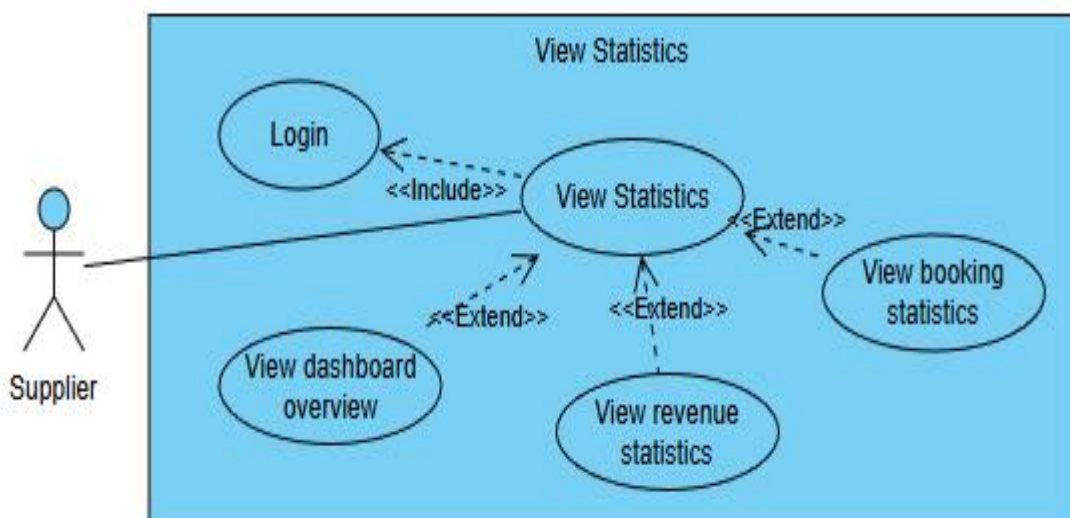


Figure 2.13. Use case View Statistics

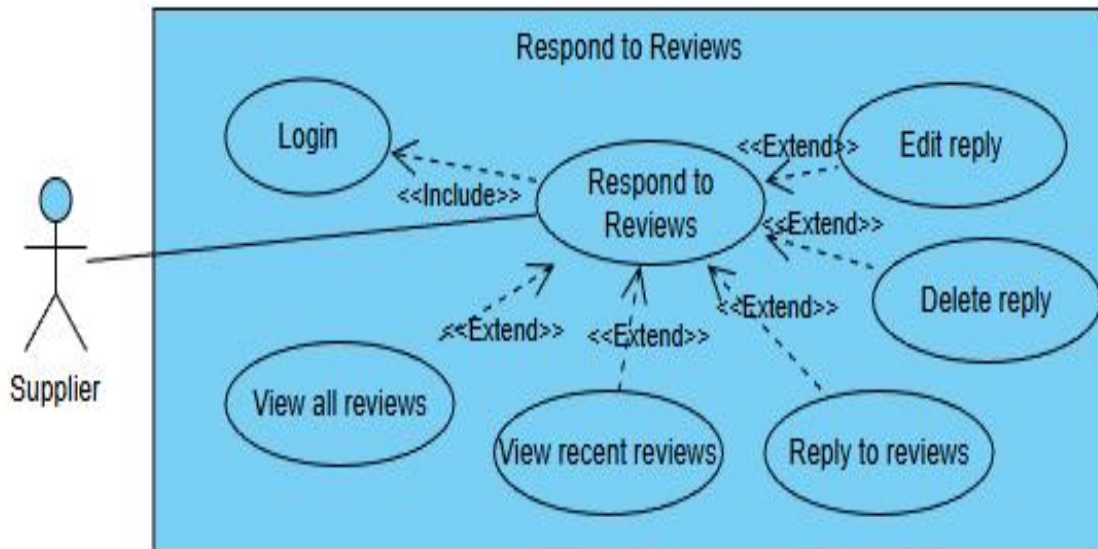


Figure 2.14. Use case Respond to Reviews

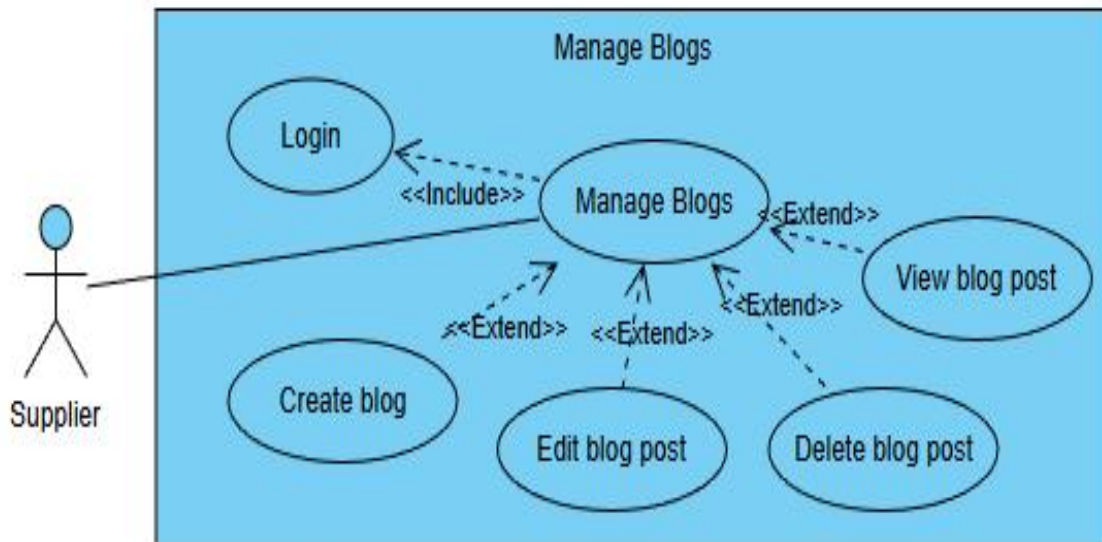


Figure 2.15. Use case Manage Blogs

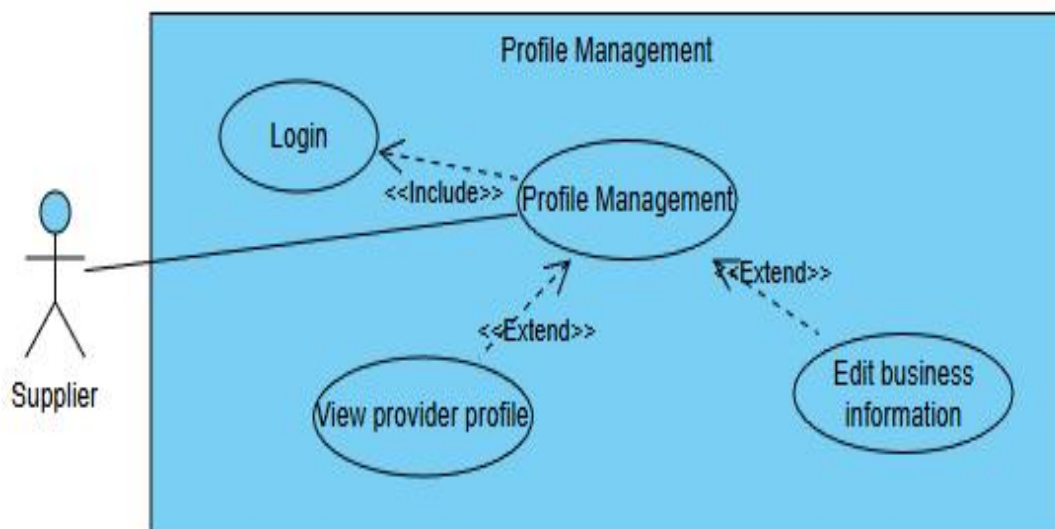


Figure 2.16. Use case Profile Management

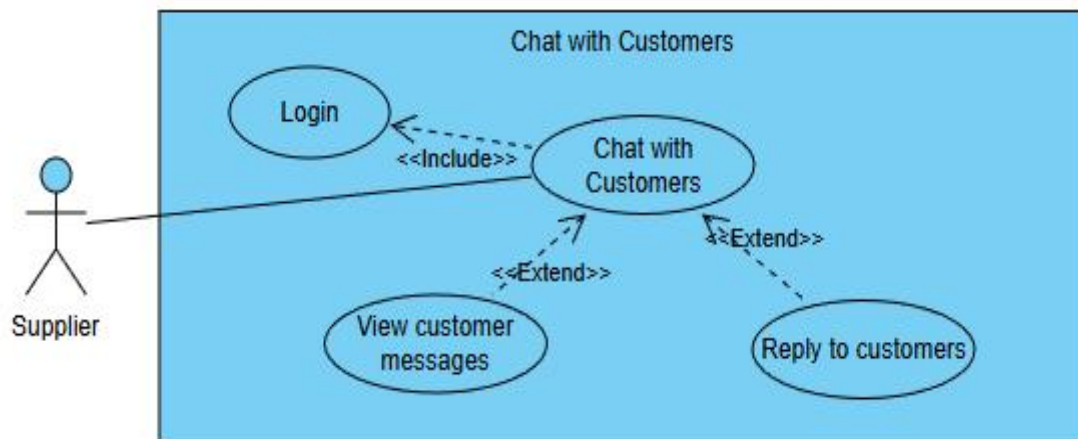


Figure 2.17. Use case Chat with Customers

2.3. ACTIVITY DIAGRAM

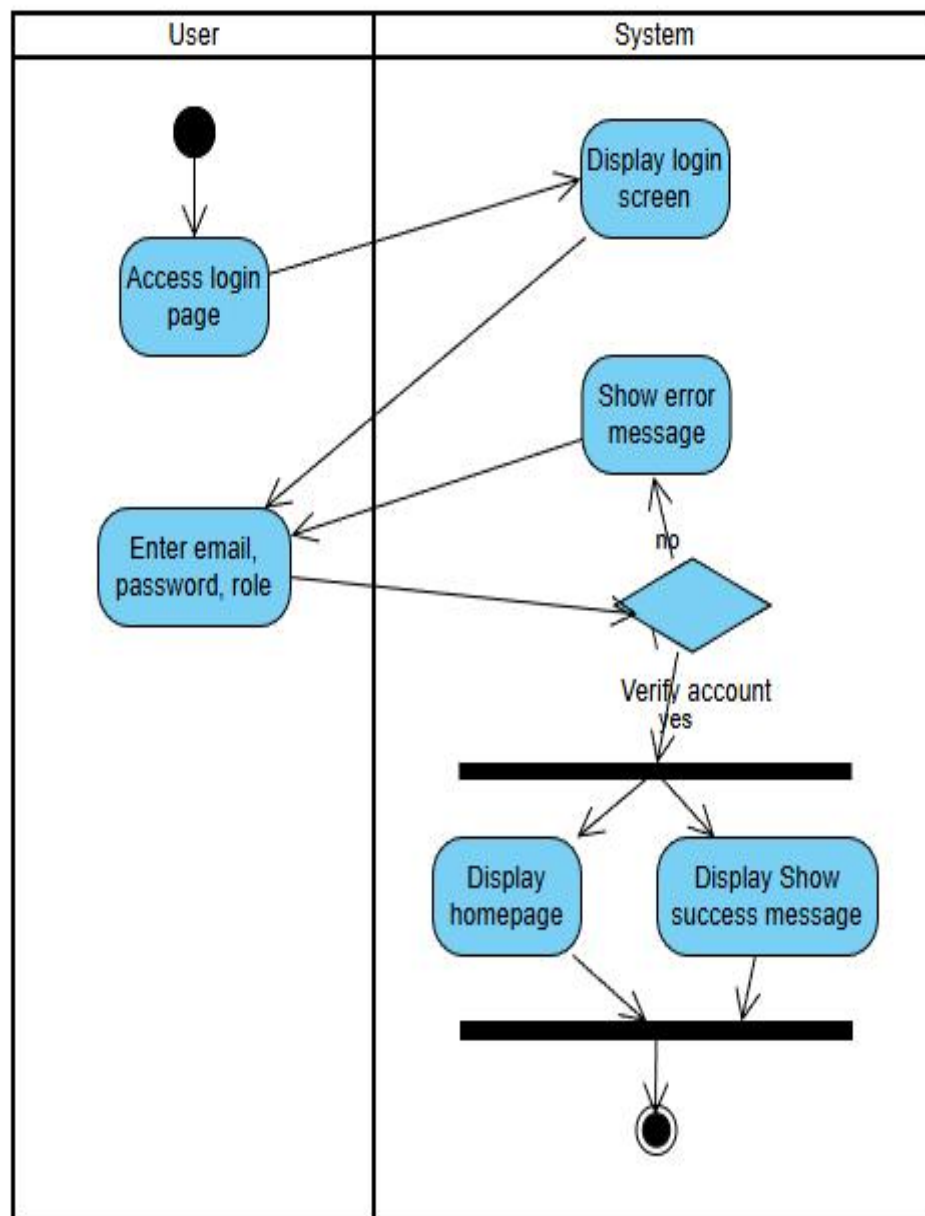


Figure 2.18. Activity diagram Login

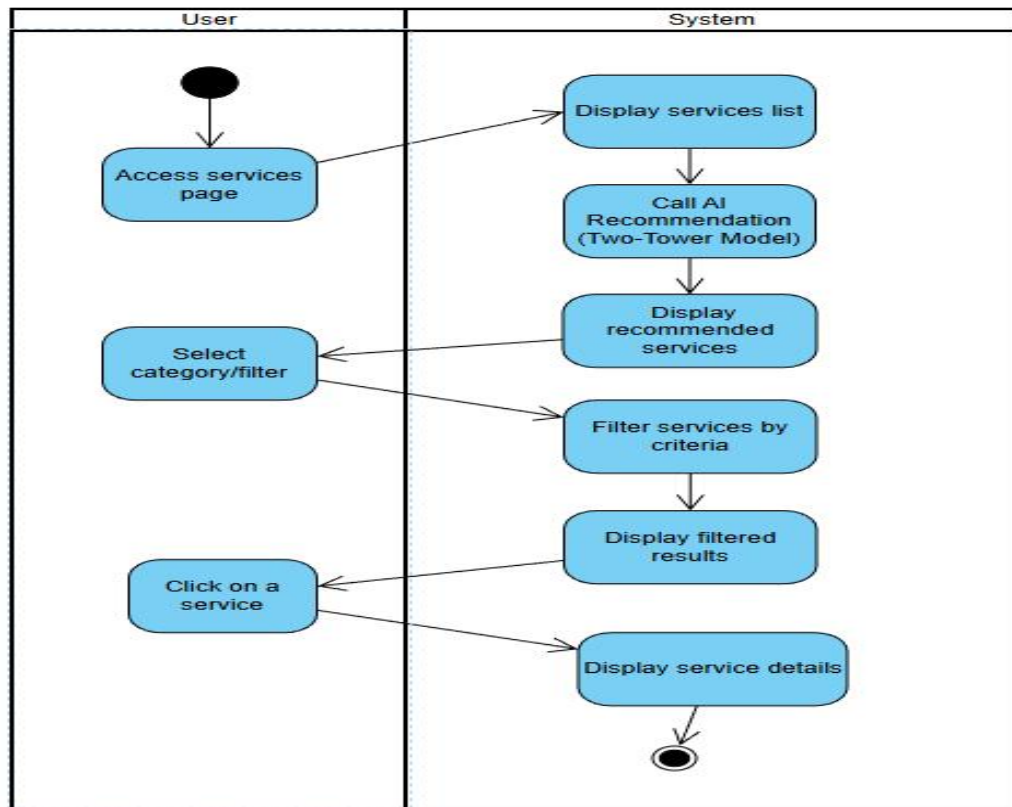


Figure 2.19. Activity diagram Browse Services

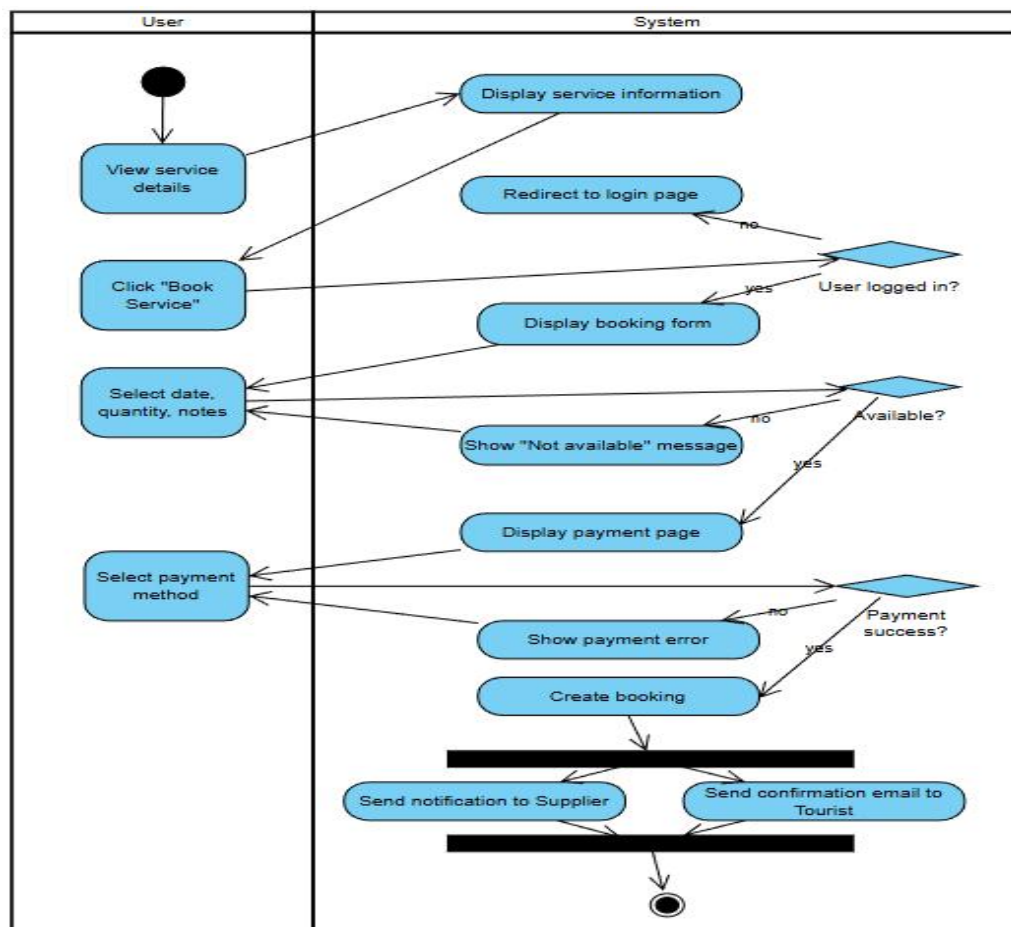


Figure 2.20. Activity diagram Book Services

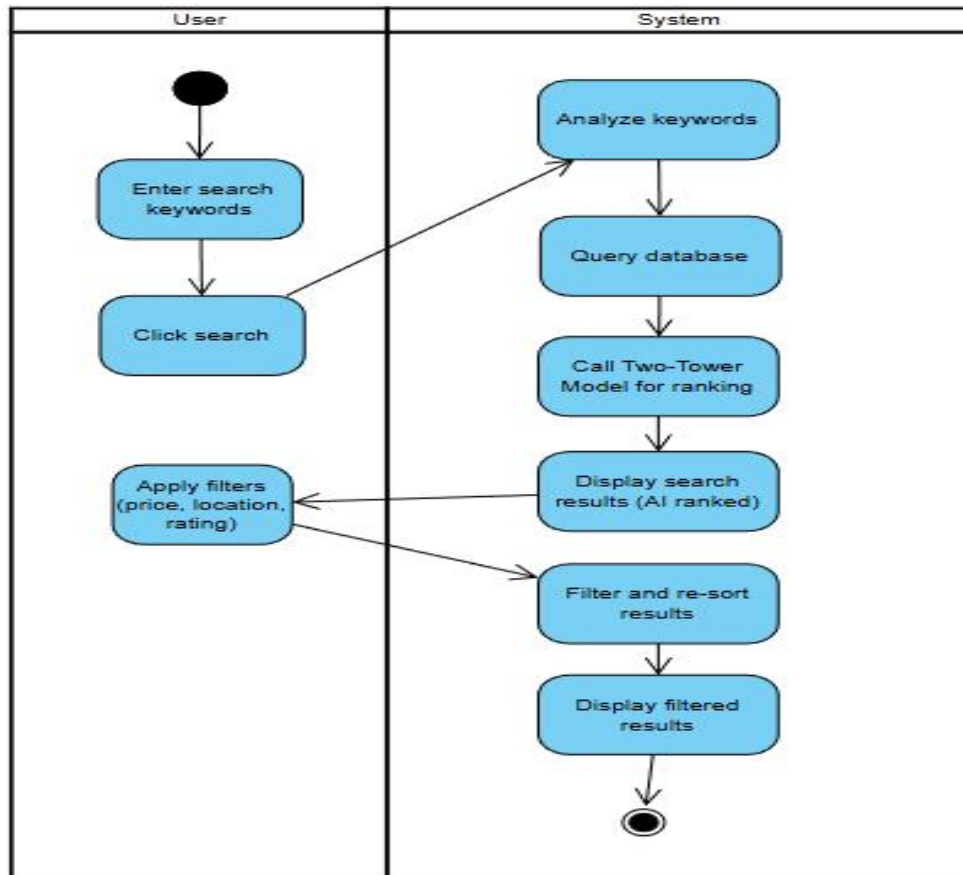


Figure 2.21. Activity diagram Search & Discovery + Ai Recommendation

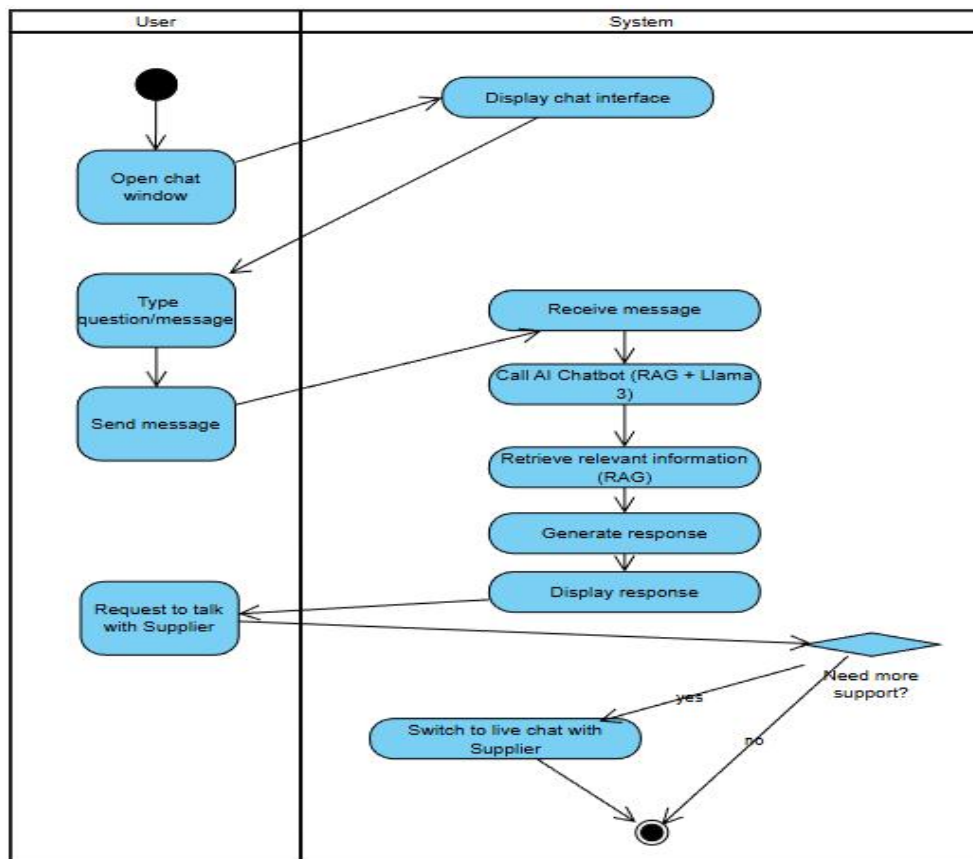


Figure 2.22. Activity diagram Chat & Support + Ai Chatbot (Rag + Llama 3)

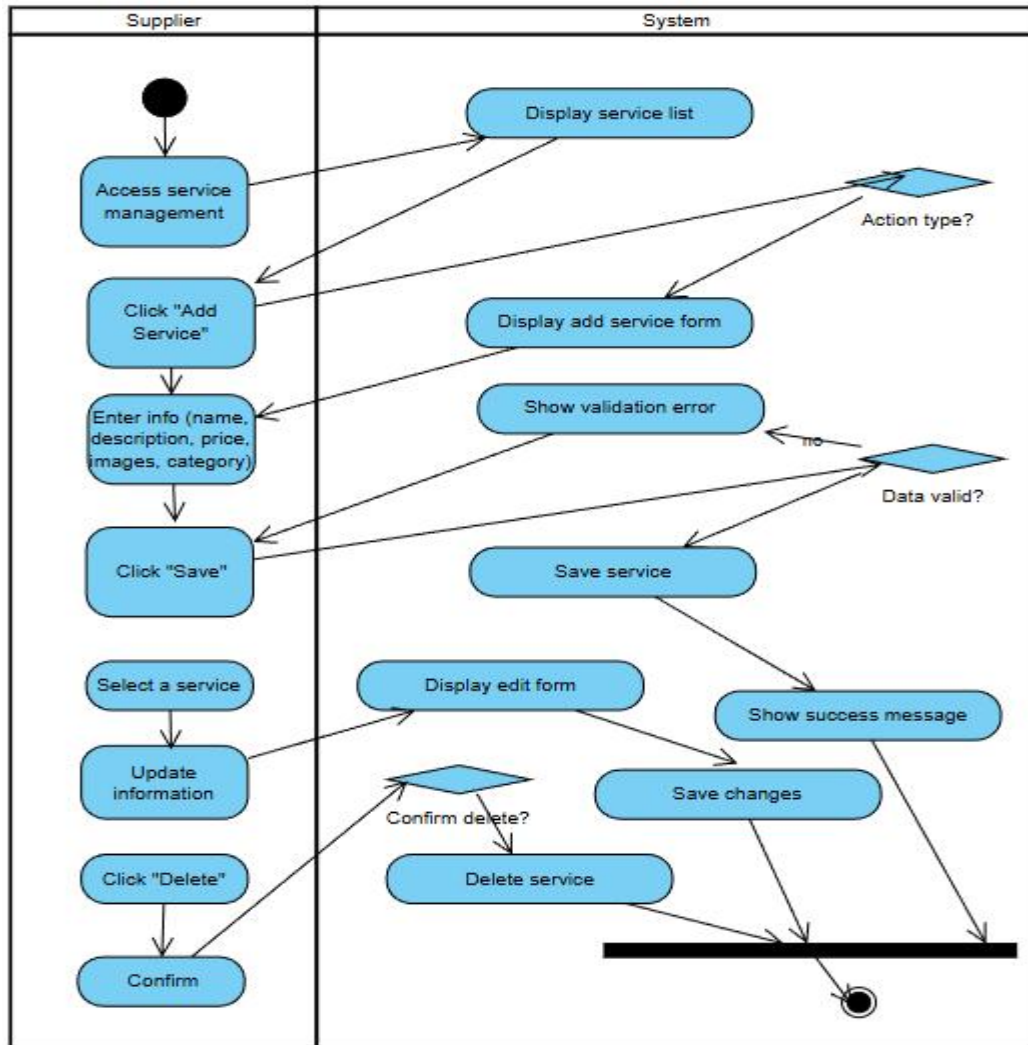


Figure 2.23. Activity diagram Manage Services

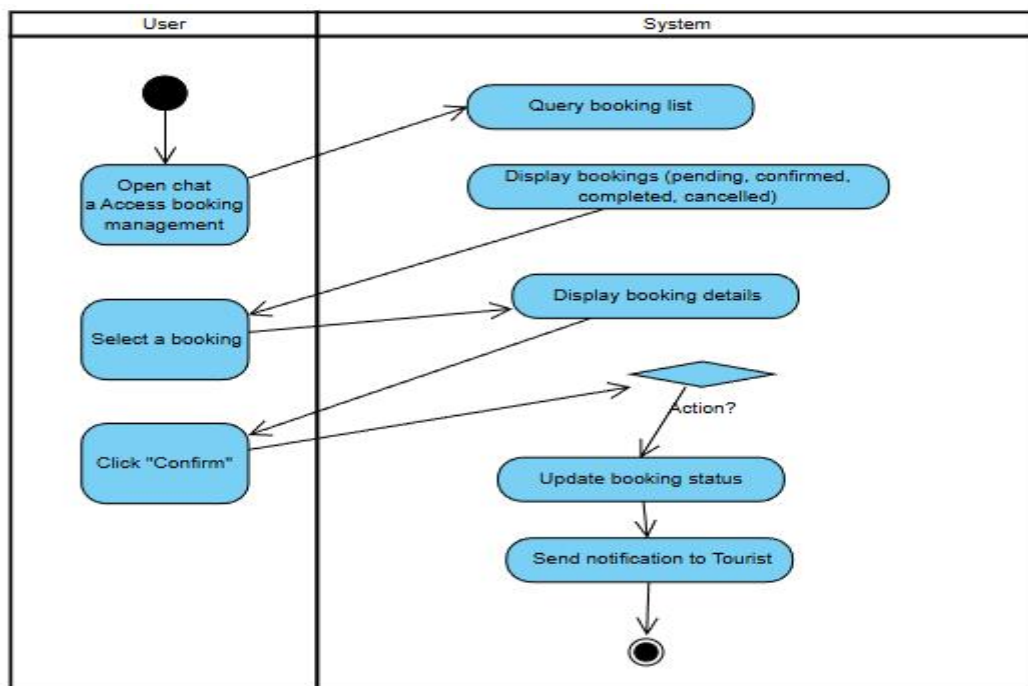


Figure 2.24. Activity diagram Manage Bookings

2.4. CLASS DIAGRAM

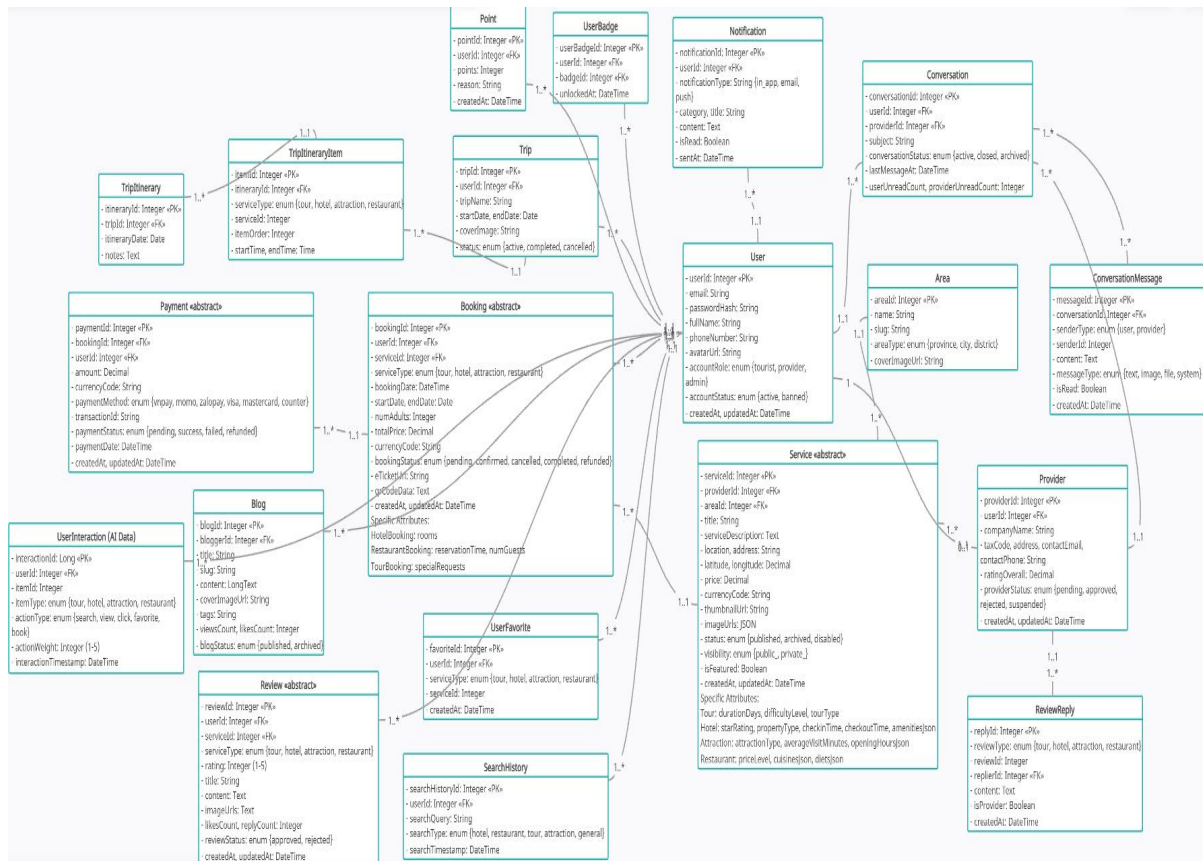


Figure 2.25. Class Diagram

2.5. SEQUENCE DIAGRAM

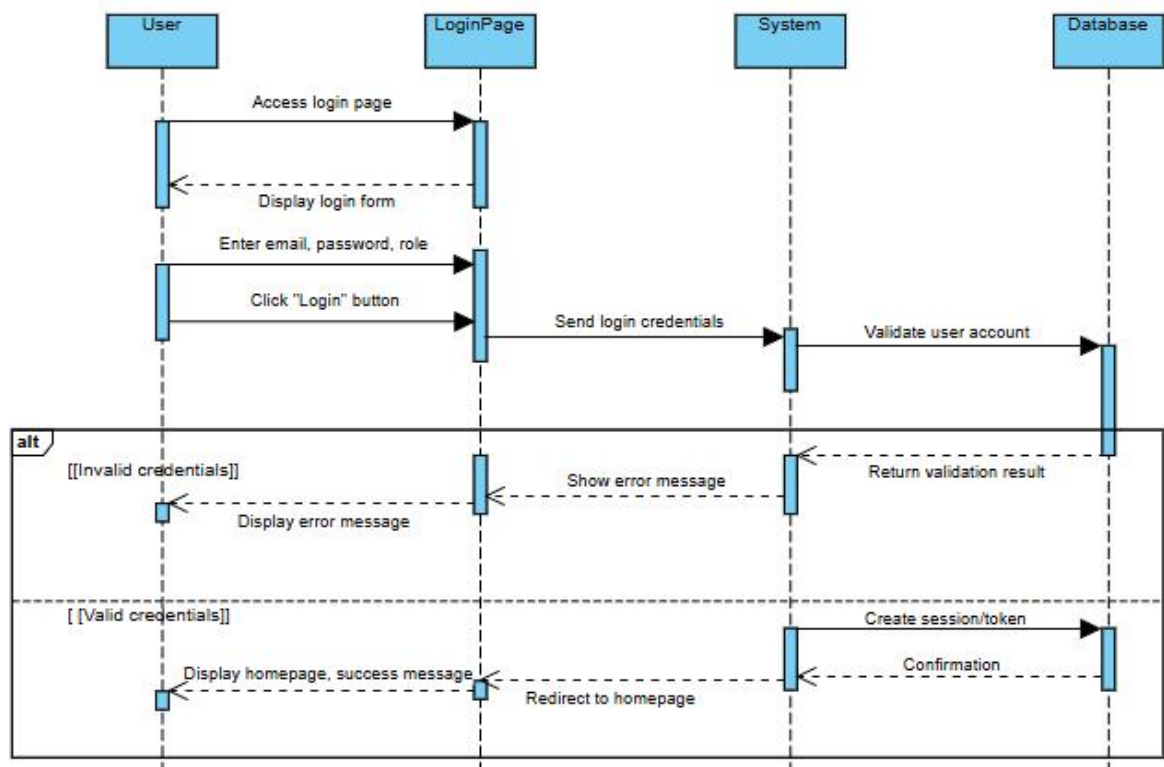


Figure 2.26. Sequence diagram Login

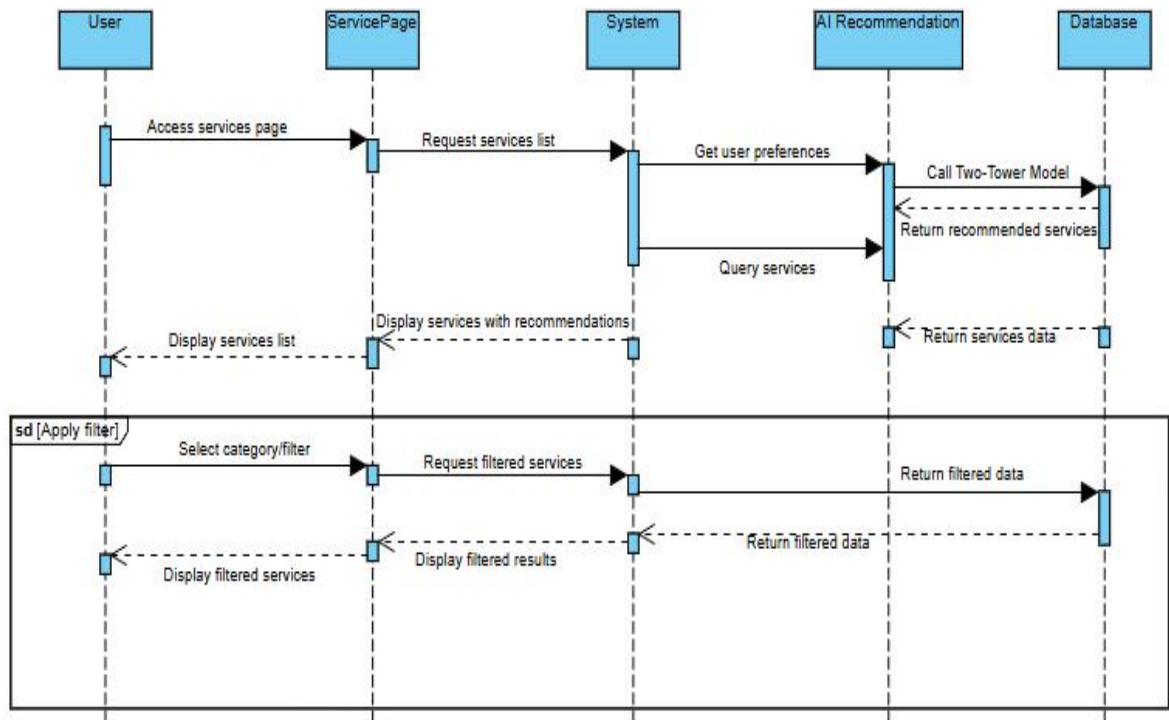


Figure 2.27. Sequence diagram Browse Services

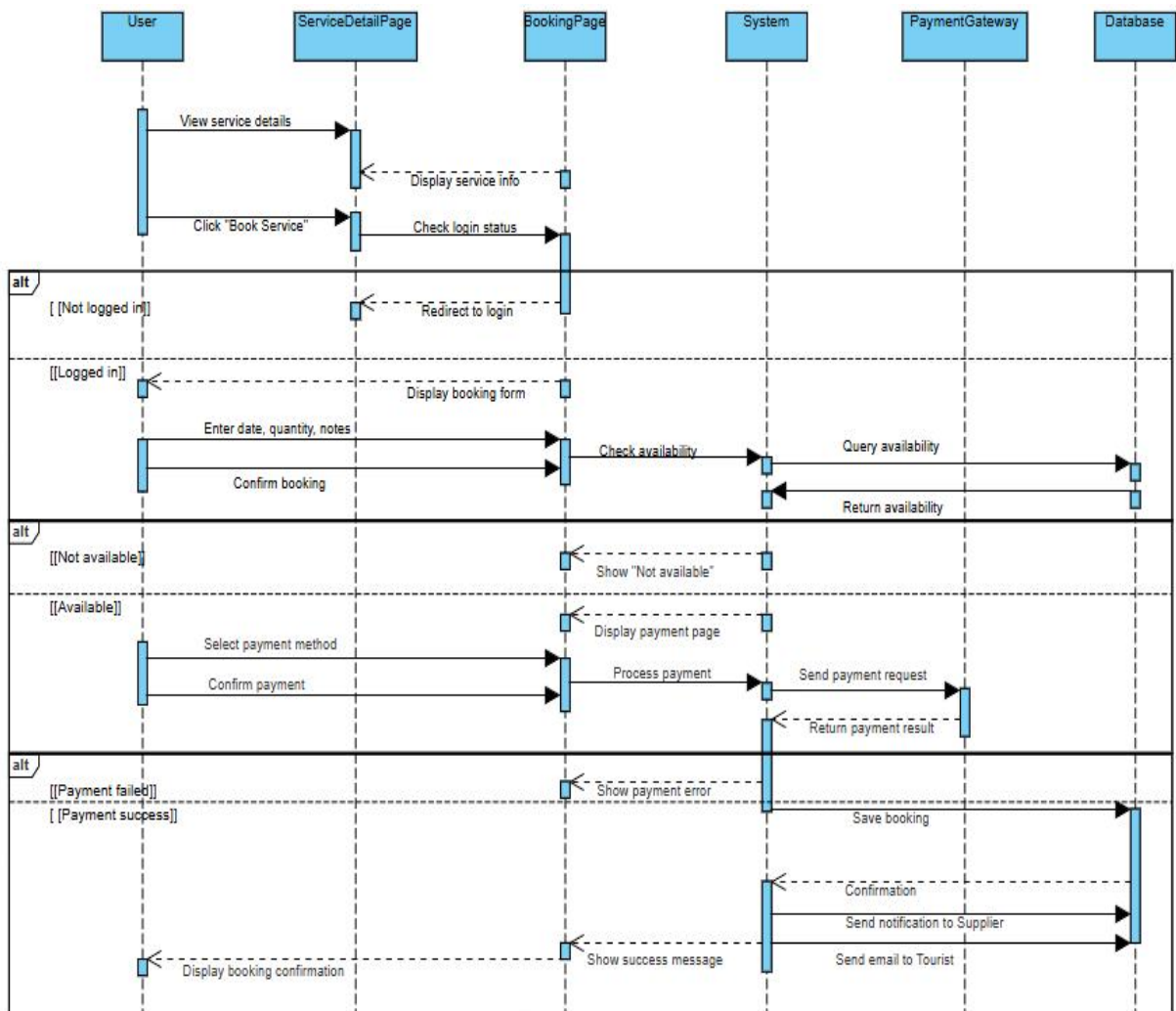


Figure 2.28. Sequence diagram Book Services

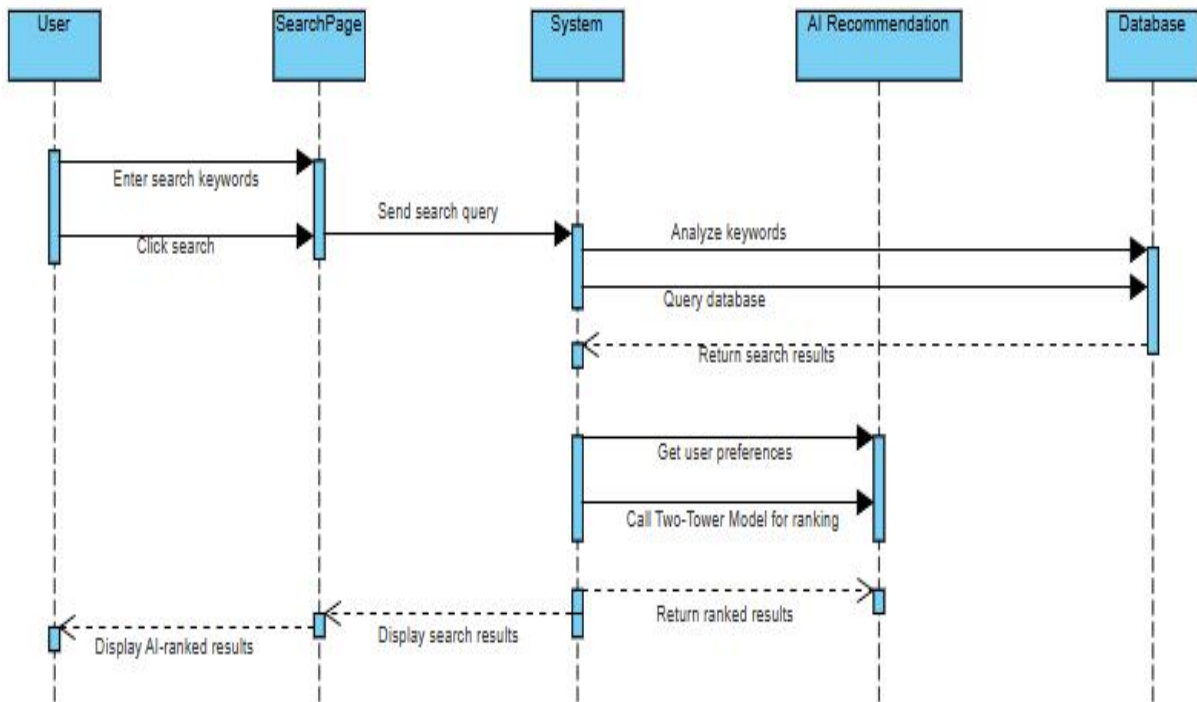


Figure 2.29. Sequence diagram Search & Discovery + Ai Recommendation

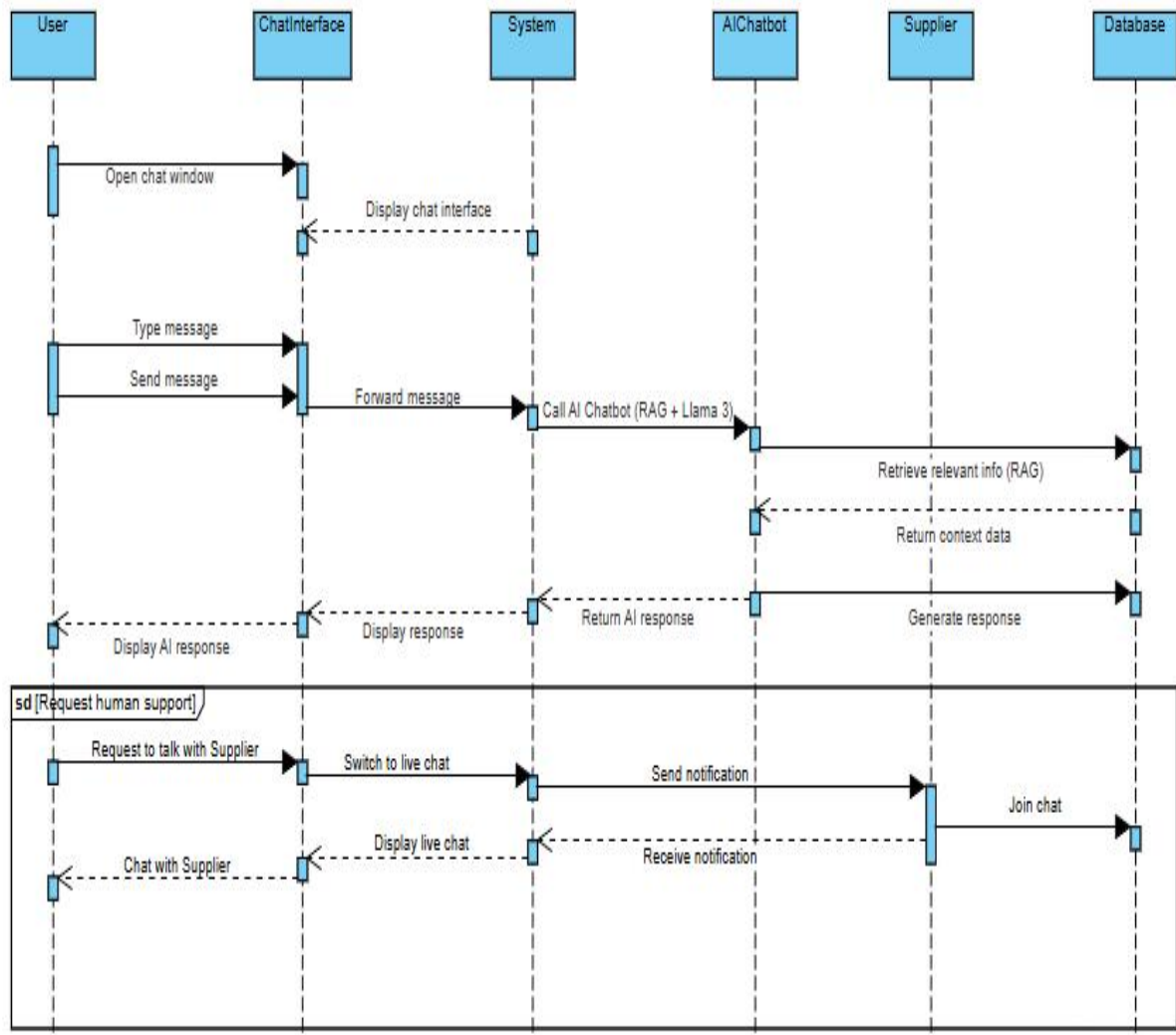


Figure 2.30. Sequence diagram Chat & Support + Ai Chatbot (Rag + Llama 3)

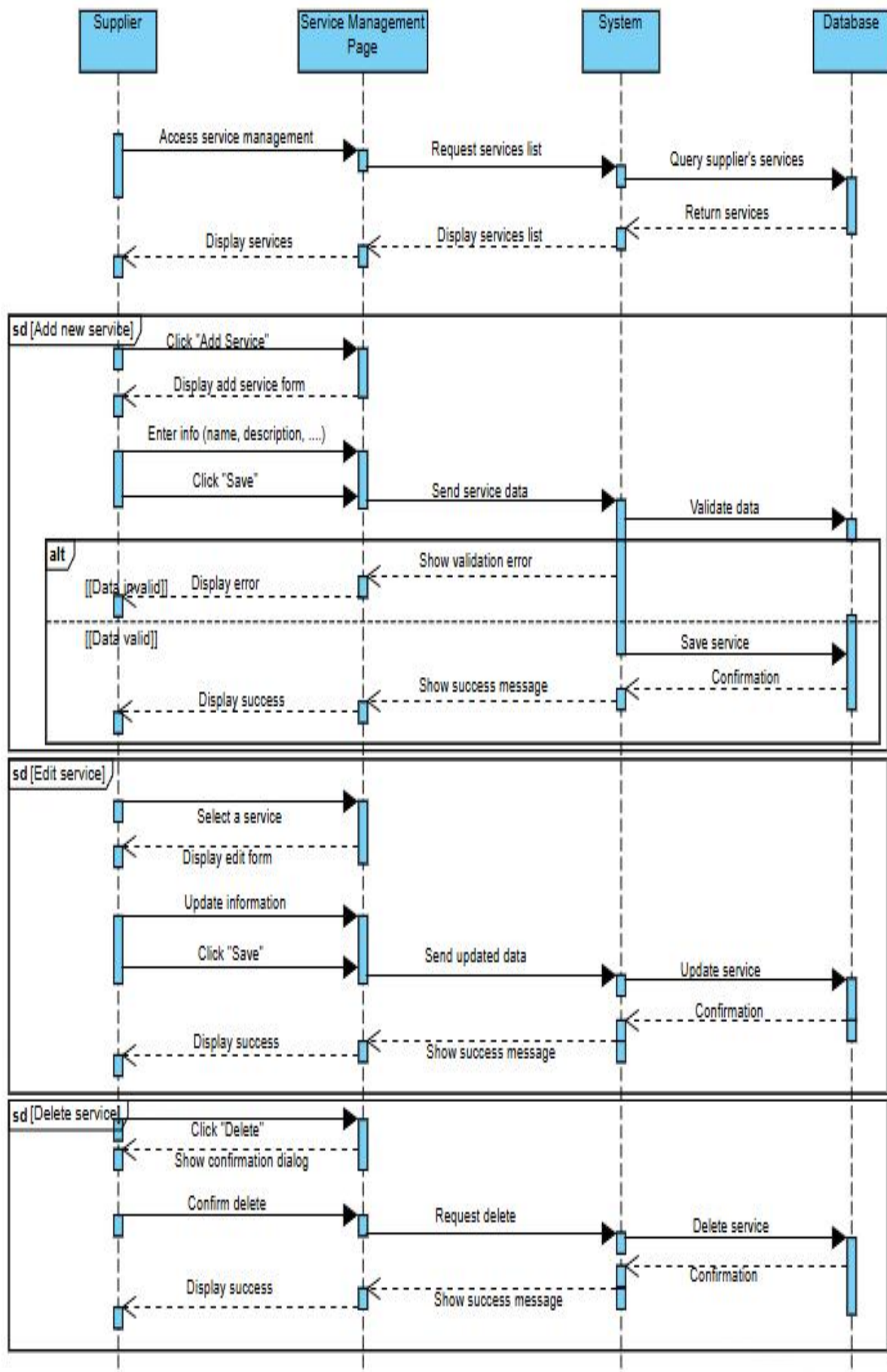


Figure 2.31. Sequence diagram Manage Services

2.6. FLOWCHART DIAGRAM

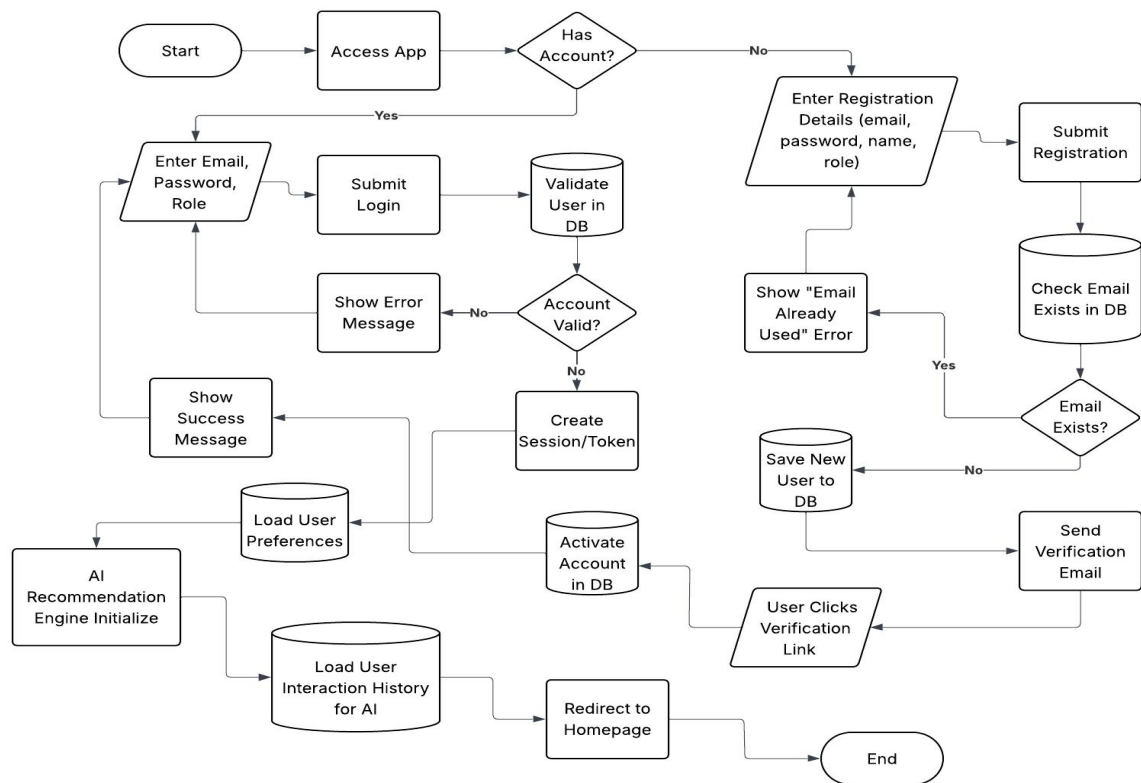


Figure 2.32. Flowchart diagram User Authentication Flow

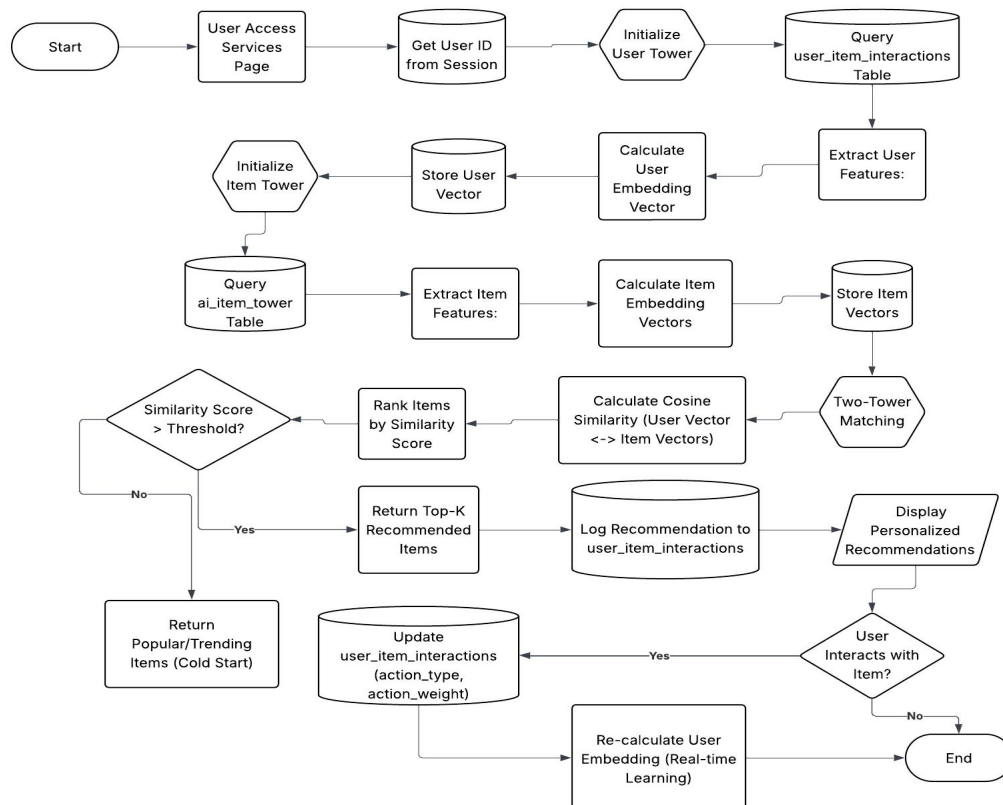


Figure 2.33. Flowchart diagram Ai Recommendation System Flow (Two-Tower Model)

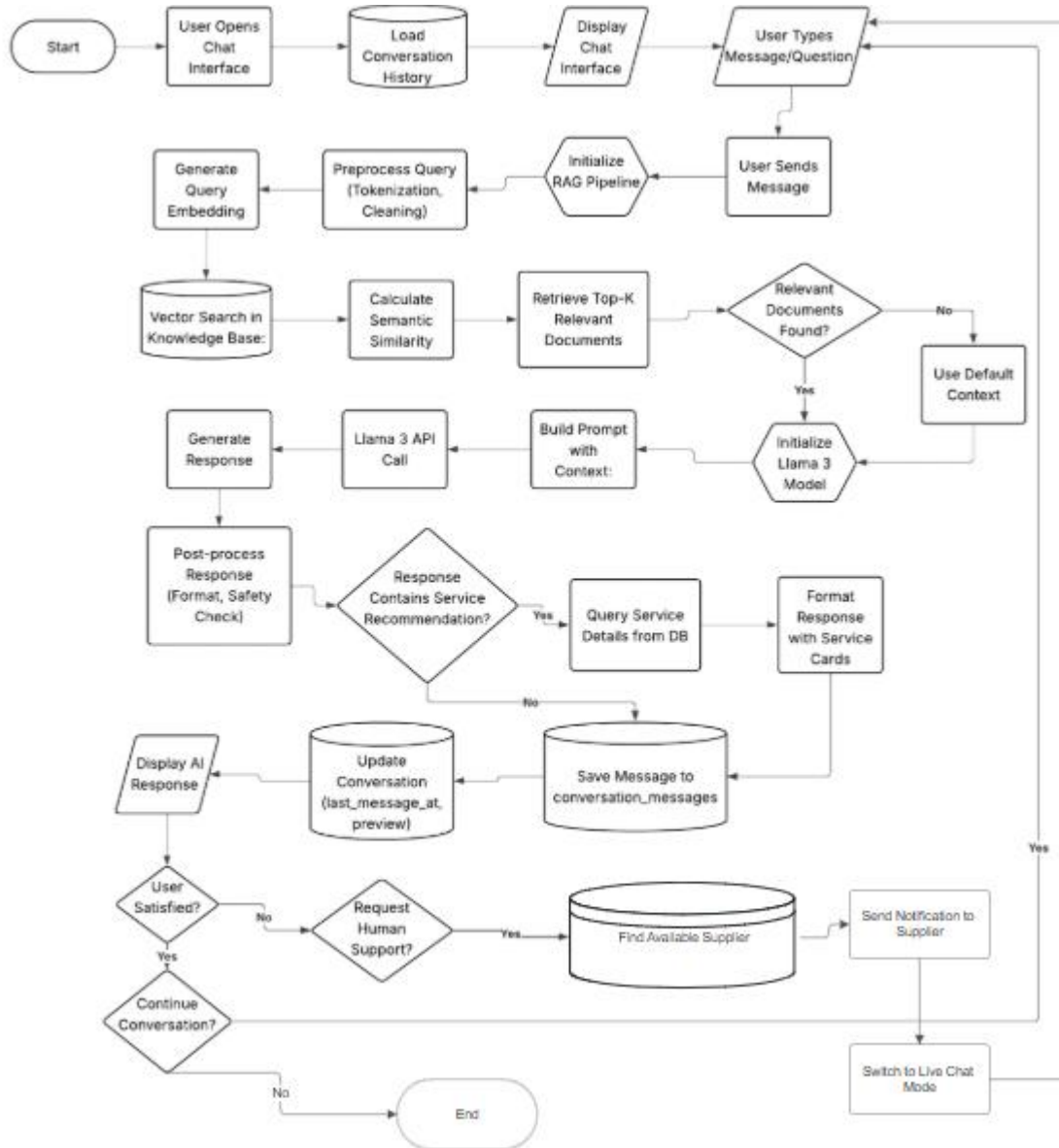


Figure 2.34. Flowchart diagram Rag Chatbot Flow (Rag + Llama 3)

2.7. SUMMARY

This chapter presents the comprehensive system analysis and design for Tripfinity - a smart travel application. The system is designed to serve two main actors: Tourist (mobile app users) and Supplier (web portal users), with 15 core use cases covering service browsing, booking, trip planning, reviews, and AI-powered features. The design documentation includes Use Case Diagram showing actor interactions, Activity Diagrams illustrating workflow processes, Sequence Diagrams depicting object communications, Class Diagram representing database structure with 13 entity groups and their relationships, and Flowchart Diagrams demonstrating the AI Recommendation System (Two-Tower Model) and RAG Chatbot (Llama 3) integration into the application flow.

CHAPTER 3. IMPLEMENTATION OF BUILDING SYSTEM

3.1. FLUTTER MOBILE APP

3.1.1. Authentication & Onboarding

This module handles user access to the application. The onboarding screen introduces new users to app features through interactive slides highlighting key functionalities like AI recommendations, trip planning, and booking services. The login screen supports email/password authentication with JWT token management for secure sessions. The registration screen collects user information including name, email, phone number, and password with validation. The forgot password screen enables account recovery by sending OTP verification codes to registered email addresses for password reset.

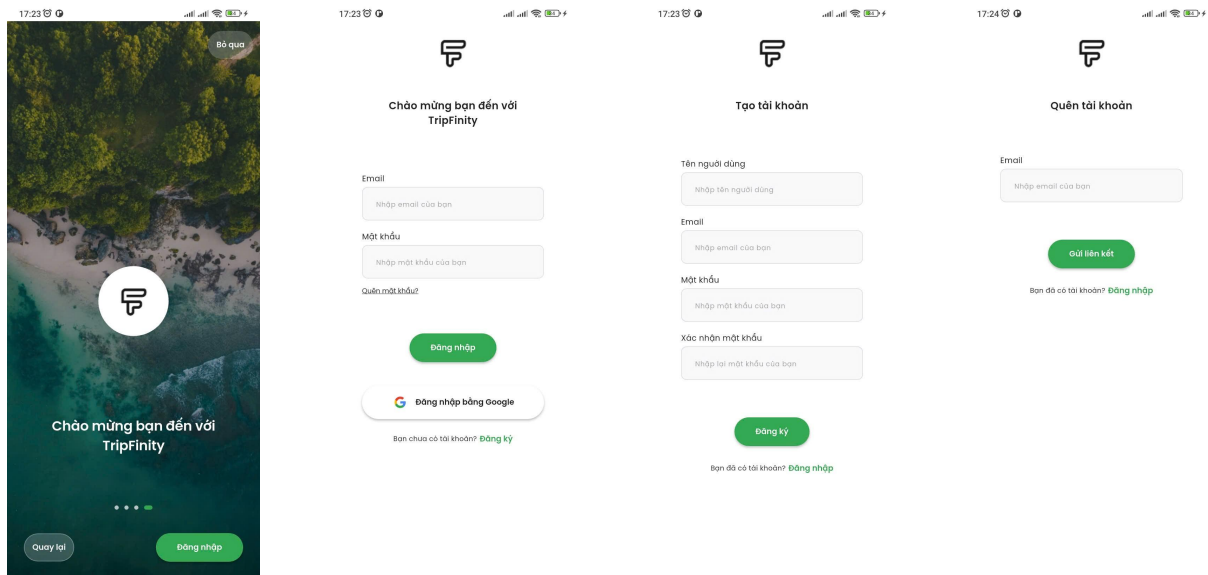


Figure 3.1. Authentication & Onboarding

3.1.2. Home & Dashboard

The home screen serves as the main entry point displaying featured services, trending destinations, and AI-personalized recommendations powered by the Two-Tower neural network model. It showcases categorized sections for tours, hotels, attractions, and restaurants with horizontal scrollable lists. The dashboard screen provides a comprehensive overview of user activity including recent bookings, favorite items, trip statistics, and quick navigation to frequently used features. The profile screen allows users to view and edit personal information, avatar, and account settings. The settings screen offers app customization options, notification preferences, language selection, and access to terms/policies and customer support contact.

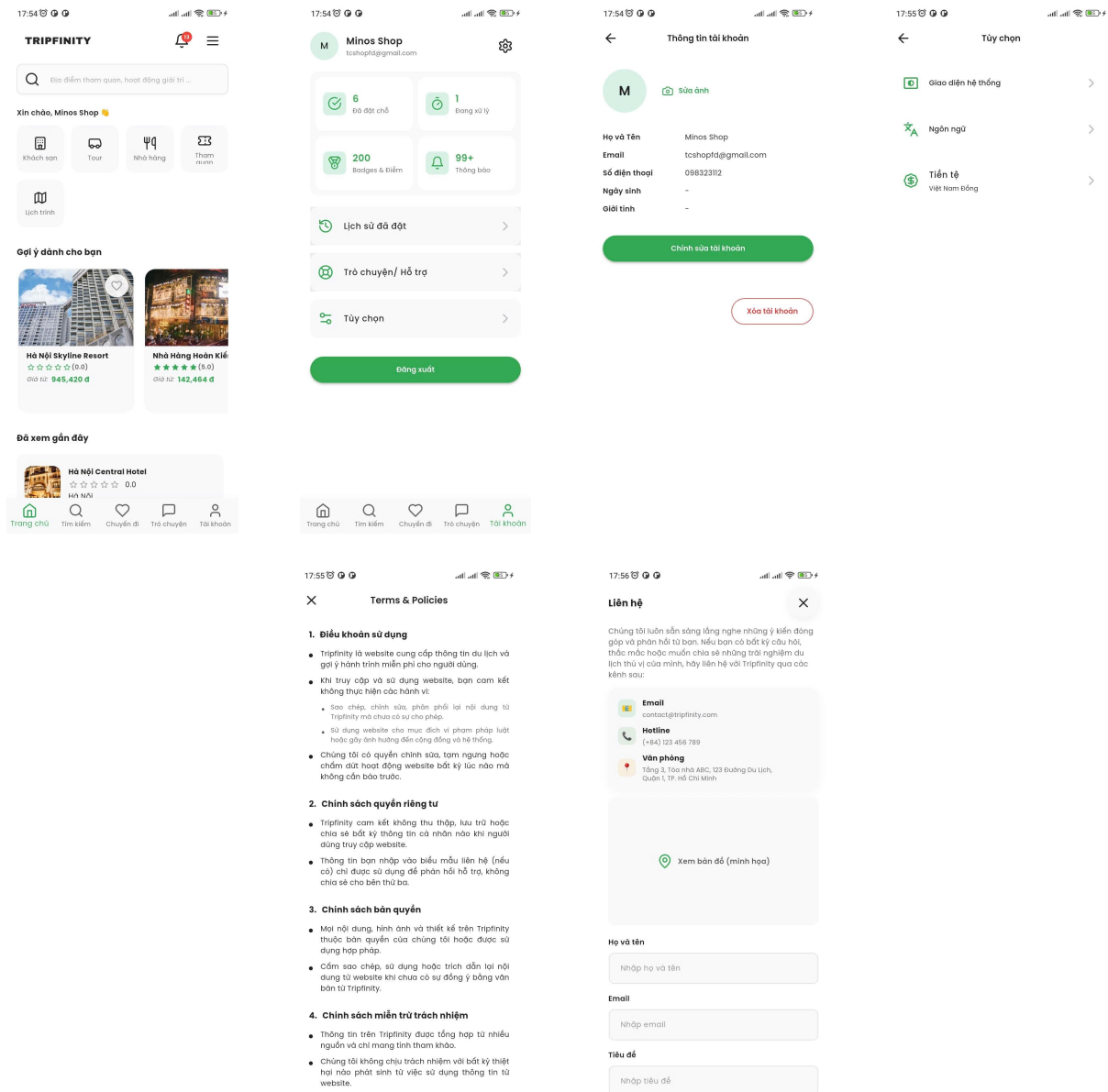


Figure 3.2. Home & Dashboard

3.1.3. Search

The general search screen provides a unified search experience across all service types with auto-suggestions and recent search history. Users can filter results by category, price range, rating, and location. The search overview screen displays categorized results with tabs for tours, hotels, attractions, and restaurants, showing item counts and relevance ranking enhanced by AI. The map search screen integrates Google Maps to display services geographically with custom markers, cluster grouping, and interactive info windows. The nearby search screen uses GPS location to find services within configurable radius, sorted by distance. The search history screen maintains a log of past searches for quick re-access and personalized suggestions.

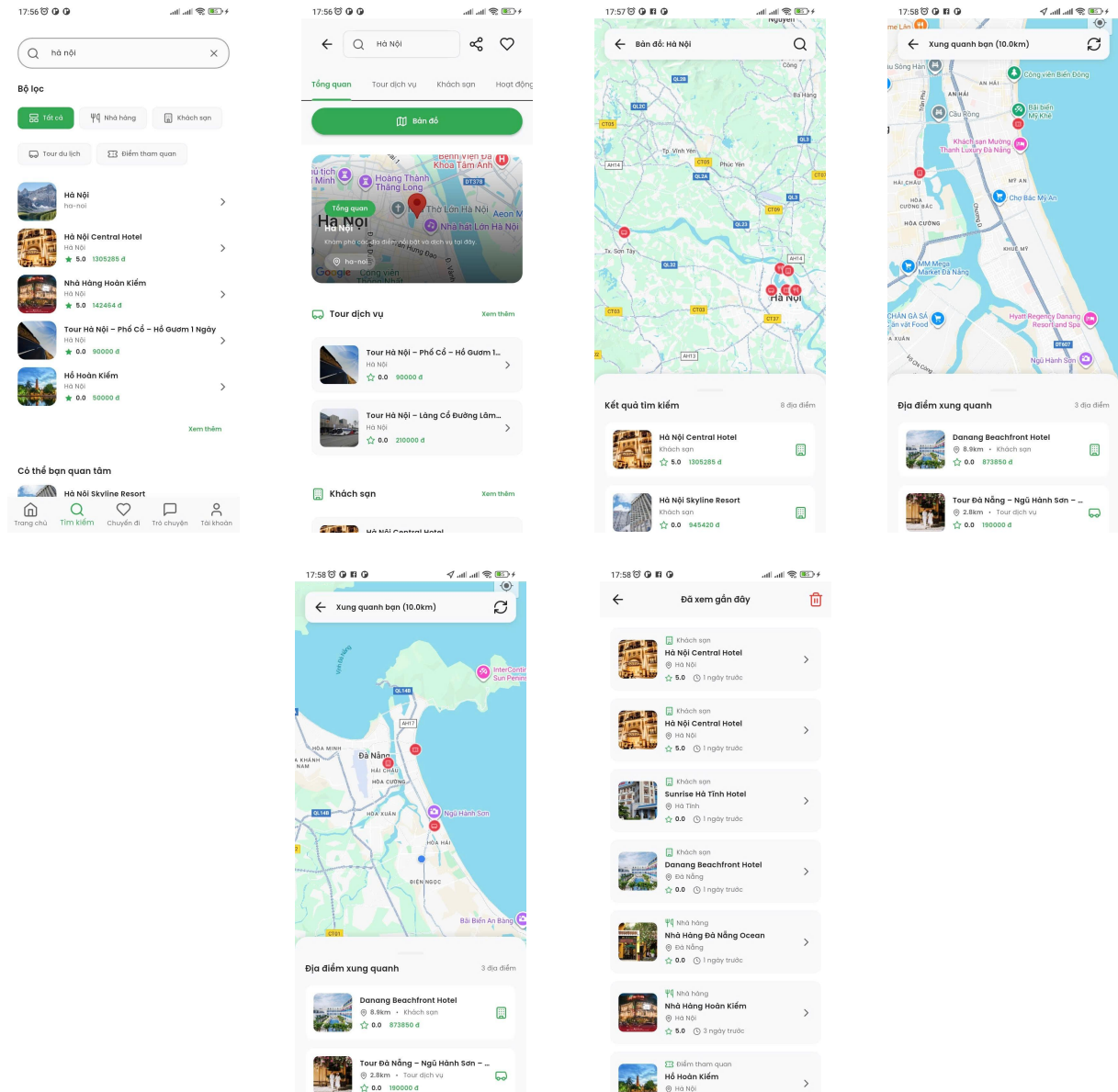


Figure 3.3. Search

3.1.4. Tour Management

The tour listing screen displays available tours with filtering options for destination, duration, difficulty level, price, and tour type (group/private/custom). Each tour card shows thumbnail, title, price, rating, and key highlights. The tour detail screen presents comprehensive information including full description, day-by-day itinerary, included/excluded services, meeting points, guide languages, difficulty level, and photo gallery with map location. The booking screen allows users to select dates, specify number of participants, add special requests, and view price breakdown. The checkout screen processes payments via ZaloPay integration or counter payment option with e-ticket generation. The reviews screen displays customer ratings with aspect-based scores (guide, value, safety, enjoyment, itinerary) and allows users to submit their own reviews with photos.

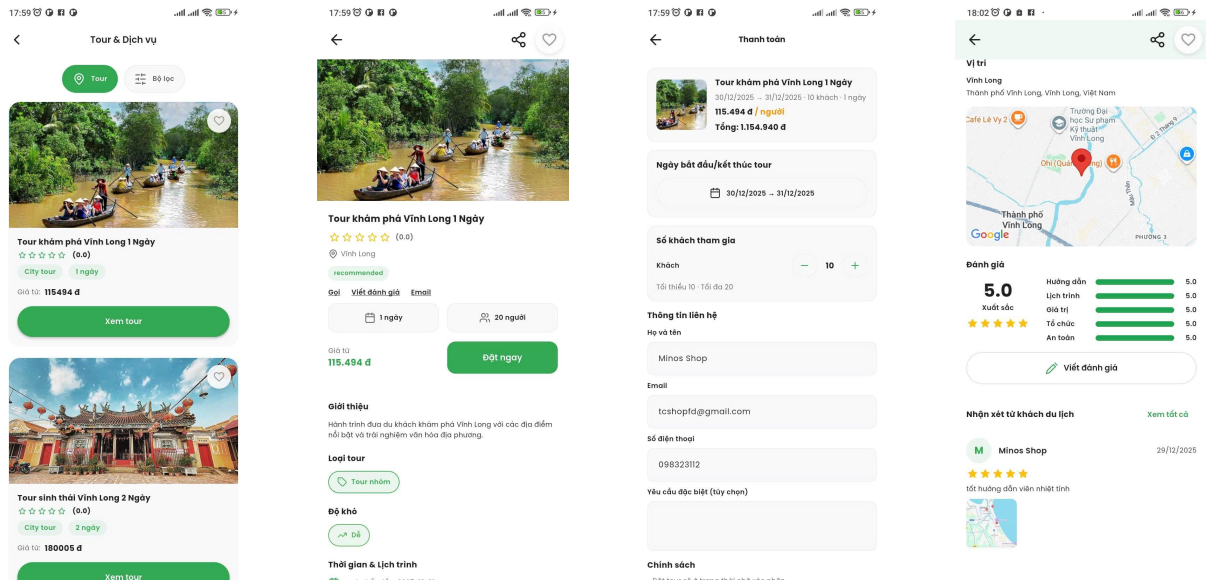


Figure 3.4. Tour Management

3.1.5. Hotel Management

The hotel listing screen showcases accommodations with filters for star rating, property type (hotel/resort/villa/apartment/hostel), price per night, amenities, and location. The hotel detail screen provides extensive information including room descriptions, check-in/check-out times, amenity icons, highlight features, policy details, and interactive map showing nearby attractions. The booking screen enables date selection with calendar, room quantity, guest count, and real-time availability checking. The checkout screen calculates total cost based on nights stayed and processes secure payments. The reviews section shows guest feedback with aspect ratings (cleanliness, service, value, location, amenities) and photo attachments.

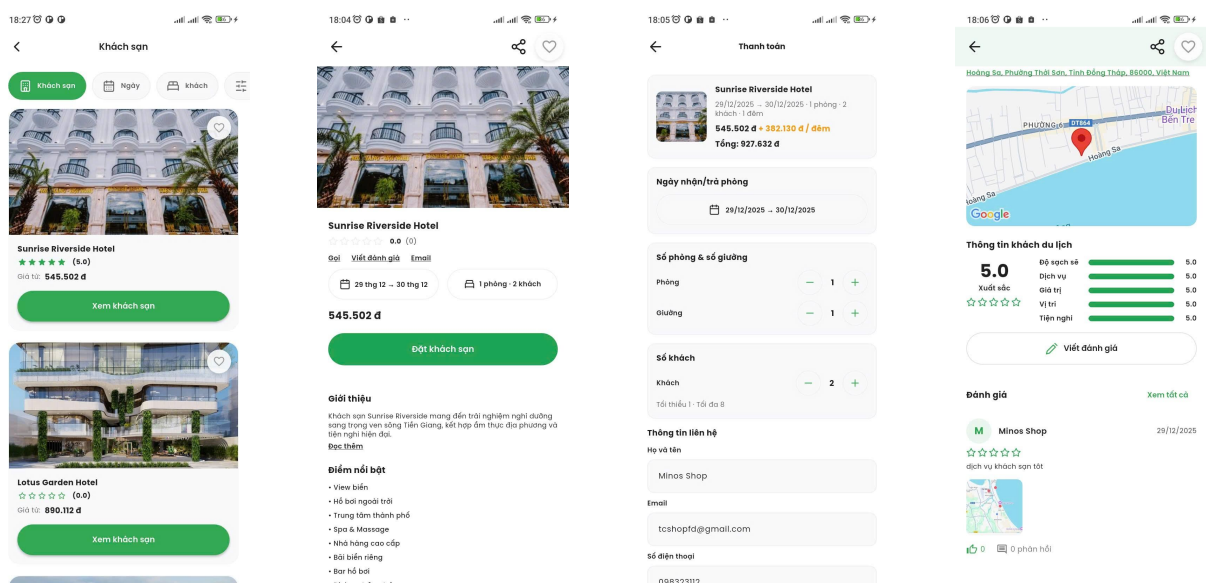


Figure 3.5. Hotel Management

3.1.6. Attraction Management

The attraction listing screen presents tourist destinations filtered by type (cultural, entertainment, historical, natural, museum, park, temple, theme park), suitable audience (family, kids, couples, solo), and price range. The detail screen displays opening hours schedule, average visit duration, visit types (guided/self-guided/audio), accessibility information, tips for visitors, and embedded map with directions. The booking screen allows date and ticket quantity selection with capacity availability. The reviews section features visitor experiences with ratings on scenery, culture, accessibility, value, and facilities.

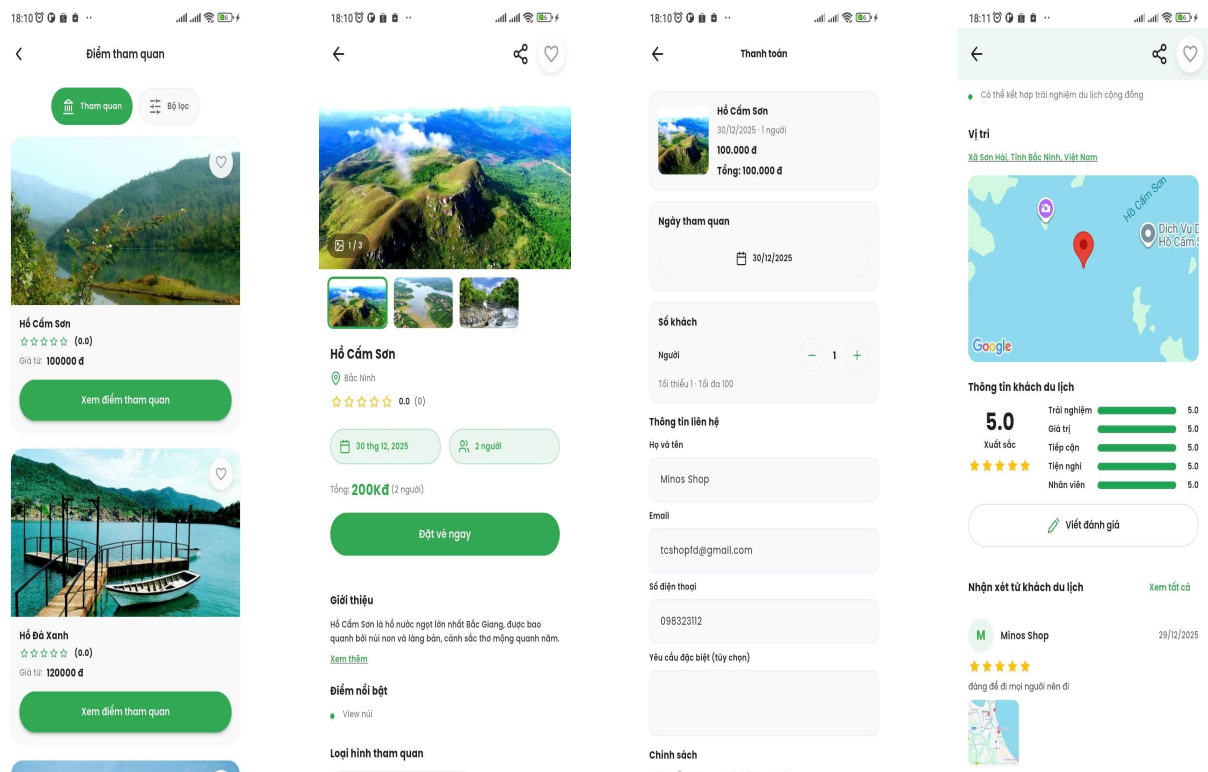


Figure 3.6. Attraction Management

3.1.7. Restaurant Management

The restaurant listing screen offers dining options filtered by cuisine type (Vietnamese, Asian, Western, Seafood, Vegetarian), dietary accommodations (halal, vegan, gluten-free), price level, and ambiance tags (romantic, family-friendly, casual, formal). The detail screen presents menu highlights, operating hours by day, reservation policies, contact information, payment methods accepted, and location map. The booking screen enables table reservation with date, time slot, party size, and special requests. The reviews section displays dining experiences with ratings on food quality, service, atmosphere, and value.

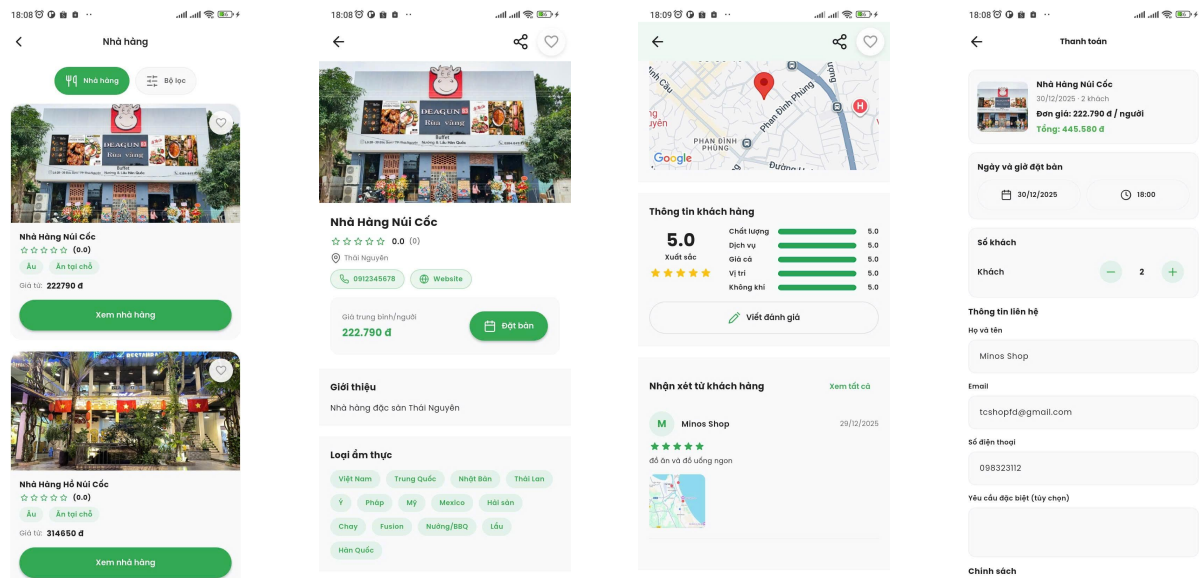


Figure 3.7. Restaurant Management

3.1.8. Booking Management

The pending bookings screen lists all reservations awaiting confirmation from suppliers, showing service details, booking date, and expected response time. The confirmed bookings screen displays approved reservations with e-ticket access, QR codes for check-in, and countdown to service date. Users can view complete booking history including completed trips and cancelled orders. Each booking card shows service thumbnail, dates, price paid, and current status. The payment webview screen handles ZaloPay transaction processing with secure redirect and callback confirmation.

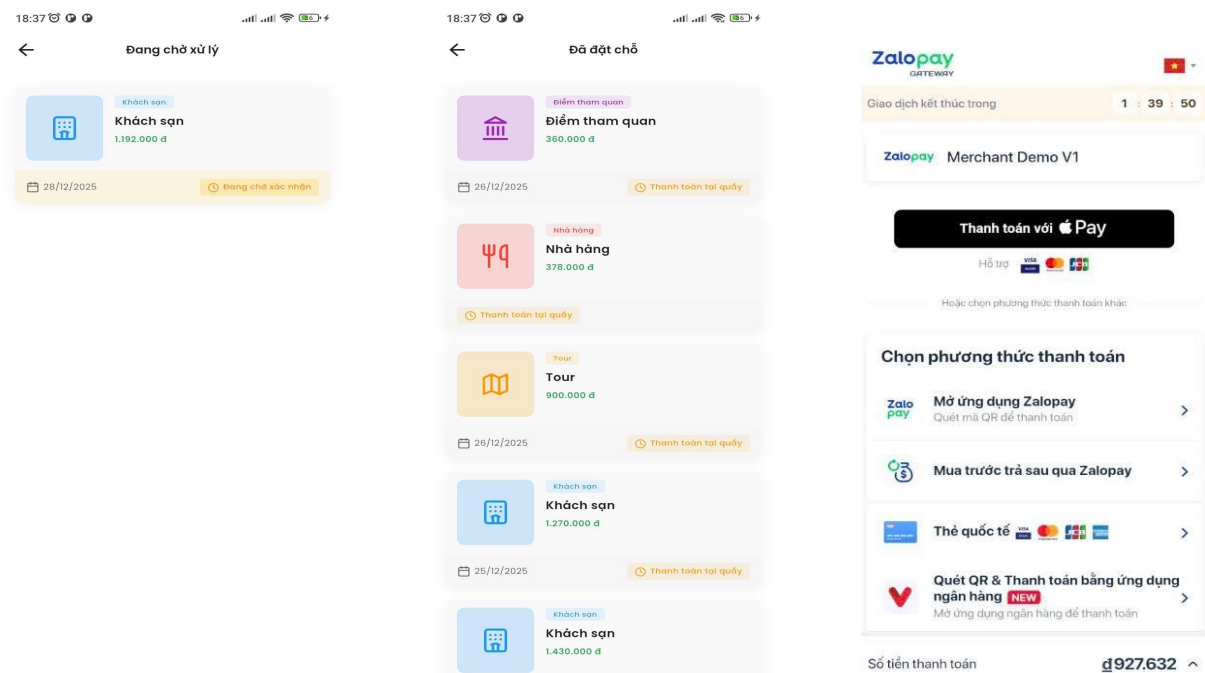


Figure 3.8. Booking Management

3.1.9. Trip Planning

The trip list screen displays user-created travel plans categorized by status (active, completed, cancelled) with cover images and date ranges for quick identification. The create trip screen allows users to name trips, set start and end dates, and upload personalized cover photos. The trip detail screen presents day-by-day itineraries with a clear timeline visualization. Users can add services such as tours, hotels, attractions, and restaurants to specific days, assign time slots, and arrange the order flexibly. The itinerary view shows daily schedules with service cards, precise timings, and map-based route visualization to support efficient planning. The trip introduction screen explains the feature benefits to first-time users.

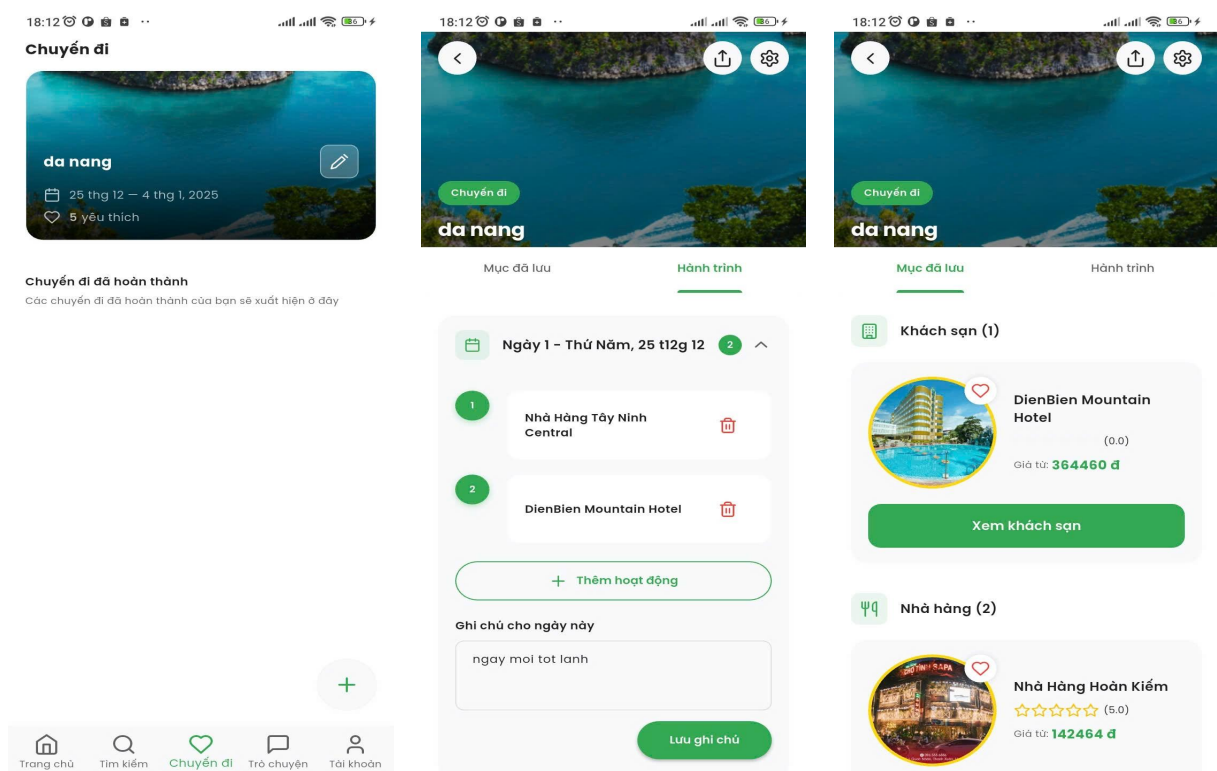


Figure 3.9. Trip Planning

3.1.10. Blog

The blog listing screen showcases travel articles from suppliers with visually engaging cover images, clear titles, author information, view counts, and publication dates to help users quickly assess relevance and popularity. Articles are intelligently categorized by tags and destinations, enabling users to browse content efficiently based on their travel interests. The blog detail screen presents full article content with rich text formatting, embedded images, and related service recommendations that enhance inspiration and trip planning. Users can interact with articles by liking them and sharing directly to popular social platforms, increasing engagement and reach.

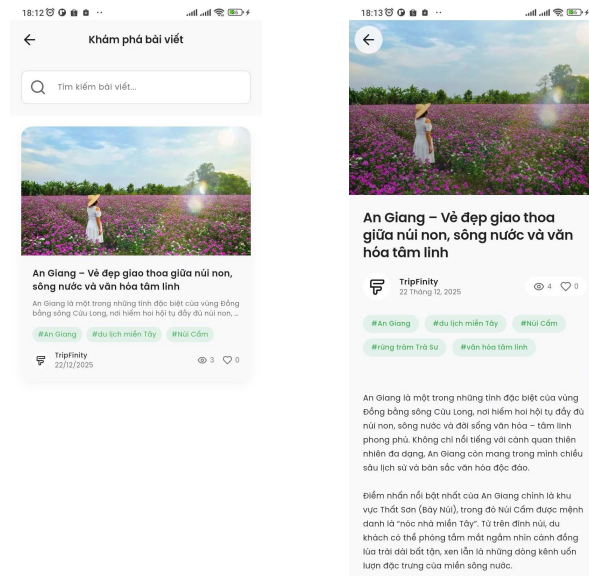


Figure 3.10. Blog

3.1.11. Chat & Support

The AI chatbot screen provides intelligent travel assistance powered by RAG (Retrieval-Augmented Generation) with Llama 3 model. Users can ask natural language questions about destinations, get personalized recommendations, compare services, and receive instant responses with relevant service suggestions. The provider chat screen enables direct messaging with service suppliers for inquiries, booking assistance, and support. The chat selection screen lists available conversations with providers showing last message preview and unread counts. Message history is preserved for reference.

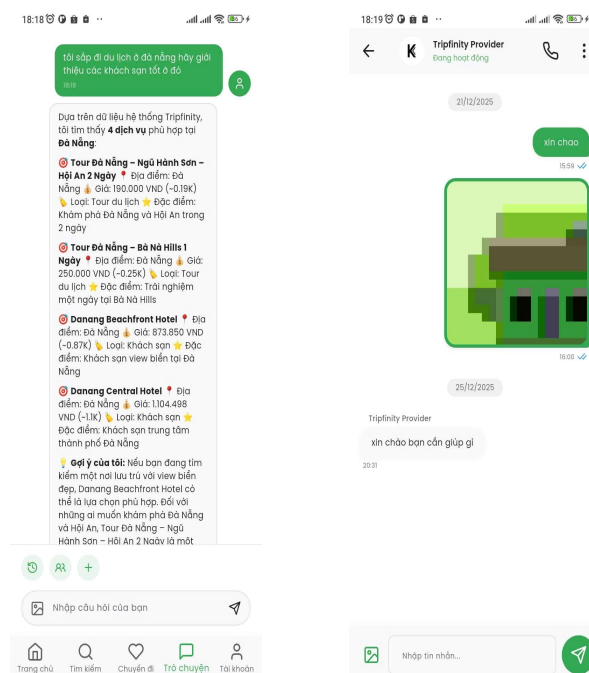


Figure 3.11. Chat & Support

3.1.12. Notifications & Rewards

The notification screen displays all system alerts including booking confirmations, payment receipts, status updates, promotional offers, and review reminders. Notifications are categorized and marked as read/unread with timestamp. The badges and points screen showcases the gamification system with earned achievement badges (first booking, review writer, frequent traveler), total accumulated points, points history with earning reasons, and progress toward next badge milestones.

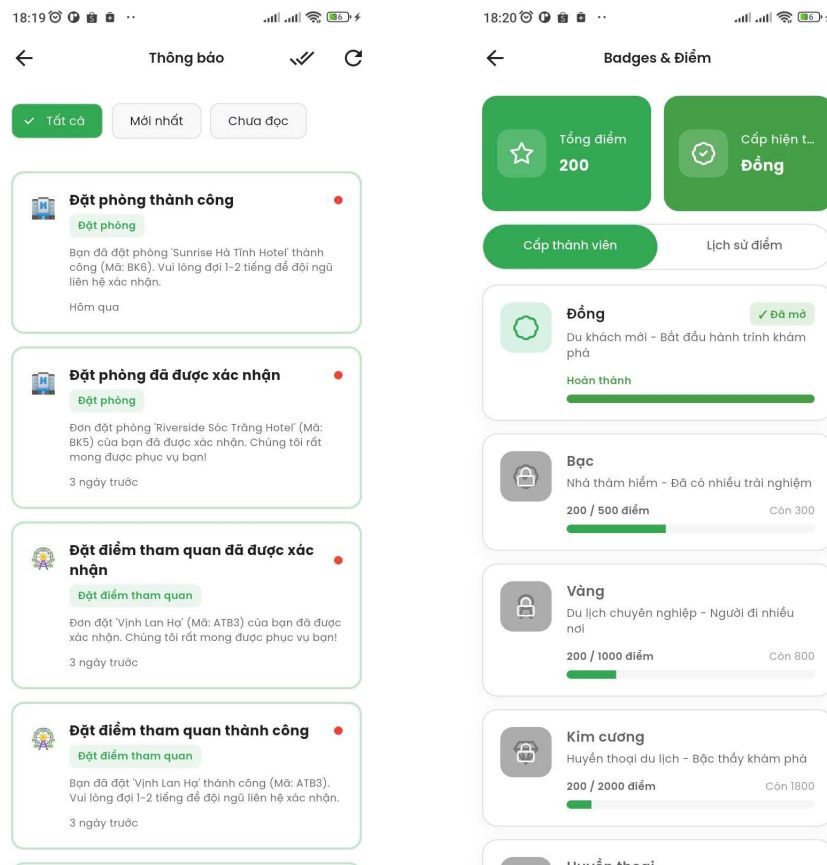


Figure 3.12. Notifications & Rewards

3.2. SUPPLIER WEB

3.2.1. Authentication

The login page provides secure access for registered suppliers with email/password authentication and session management. The registration page collects business information including company name, tax code, contact details, bank account for payments, and uploads logo and business license for verification. The forgot password page enables account recovery through email verification. The provider info page allows new suppliers to complete their business profile setup after initial registration, with status tracking for admin approval process (pending, approved, rejected, suspended).

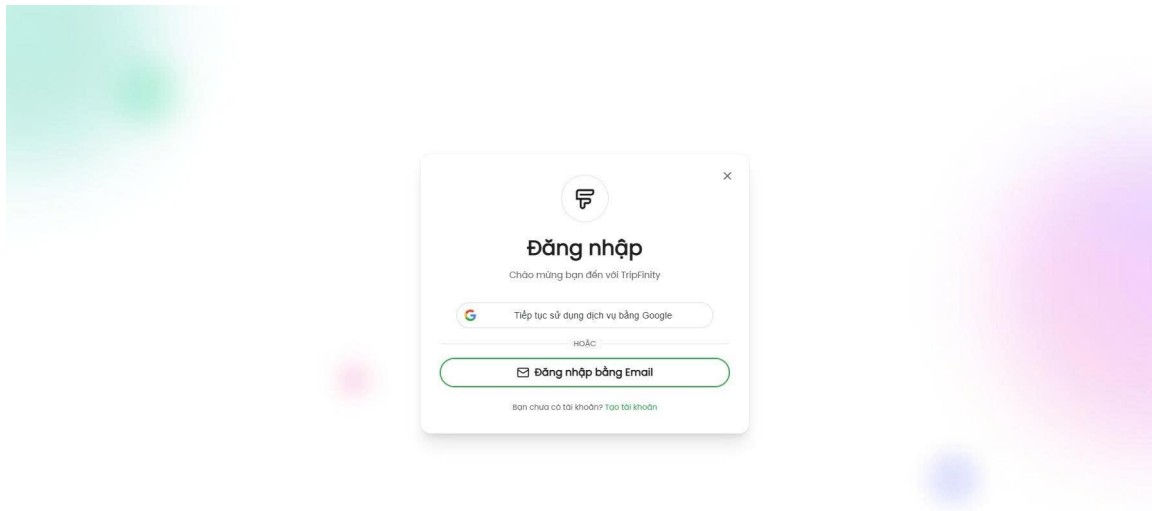


Figure 3.13. Authentication

3.2.2. Dashboard & Profile

The supplier home page presents a comprehensive business dashboard with key performance indicators including total revenue, booking counts by service type, average ratings, and recent activity timeline. Visual charts display revenue trends, booking statistics over time, and rating distribution. Quick action buttons provide shortcuts to common tasks. The profile page allows suppliers to update company information, contact details, banking information, logo, and business description. The notification page lists system alerts for new bookings, customer messages, review notifications, and payment confirmations with filtering and bulk actions. The listings page provides an overview of all services across categories with status indicators and quick navigation.

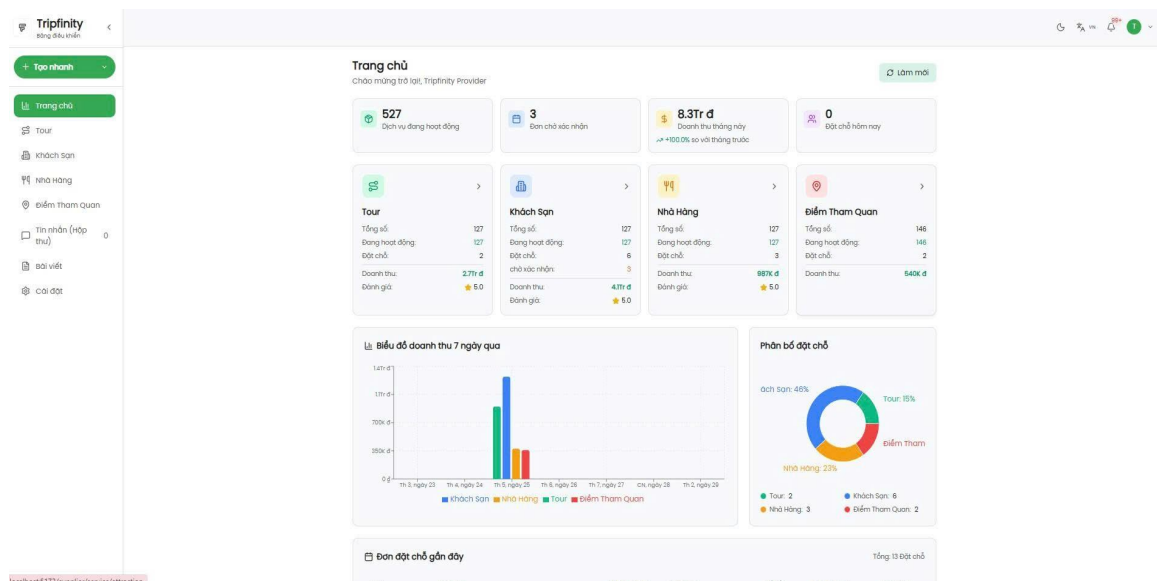


Figure 3.14. Dashboard & Profile

3.2.3. Tour Management

The tour dashboard page displays tour-specific analytics including total tours, active bookings, revenue breakdown, and performance metrics. The tour list page shows all tours with status badges (published, draft, archived), quick edit actions, and sorting/filtering options. The tour create page provides a comprehensive form for adding new tours with fields for title, description, itinerary builder, pricing, capacity, images upload, location picker with map, difficulty level, guide languages, included/excluded services, and SEO settings. The tour edit page allows modification of existing tour details with version history. The tour view page presents detailed tour information in read-only format. The booking list page displays all tour reservations with customer details, dates, participant counts, payment status, and action buttons for confirmation/cancellation. The booking detail page shows complete reservation information with customer contact and special requests. The reviews pages (all, recent, detail) enable monitoring of customer feedback, displaying ratings, comments, and providing response functionality for supplier replies.

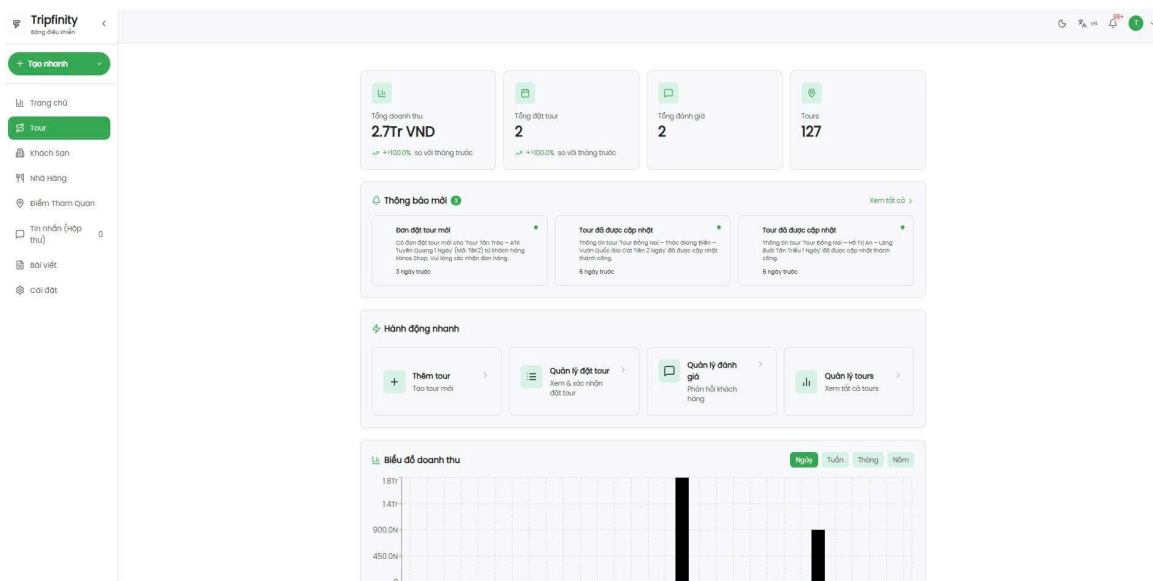


Figure 3.15. Tour Management

3.2.4. Hotel Management

The hotel dashboard page presents hotel-specific metrics including occupancy rates, room bookings, revenue per available room, and guest satisfaction scores. The hotel list page displays all properties with star ratings, room counts, and status indicators. The hotel create page offers extensive form fields for property details including title, description, star rating, property type, address with map picker, check-in/check-out times, room configurations (total rooms, beds per room), amenities selection from predefined list, highlight features, policy text, and multiple image uploads. The hotel edit page allows updating property information and availability.

The booking pages manage room reservations with guest details, stay dates, room assignments, and payment tracking. Check-in/check-out status management is available. The review pages display guest feedback with aspect ratings and enable supplier responses to address concerns or thank satisfied guests.

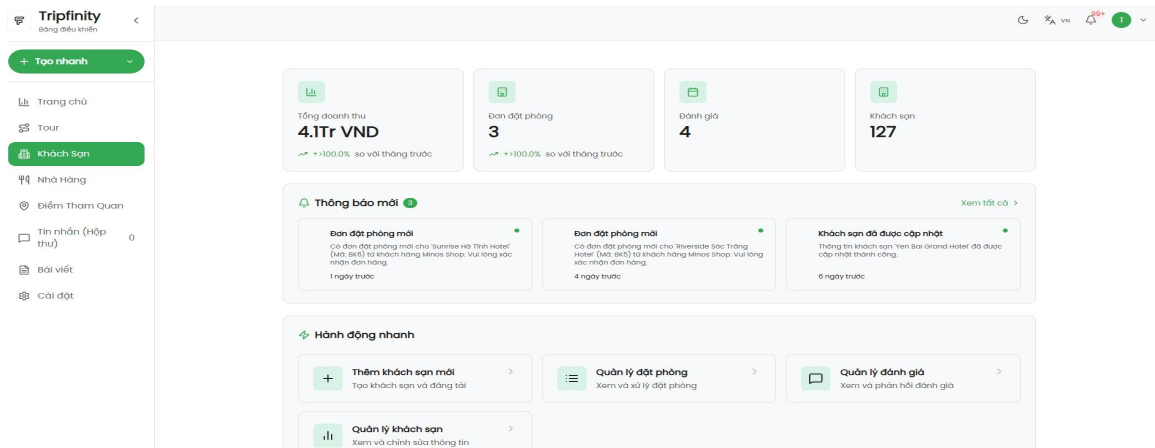


Figure 3.16. Hotel Management

3.2.5. Attraction Management

The attraction dashboard page shows venue performance data including ticket sales, visitor counts, revenue trends, and satisfaction ratings. The attraction list page displays all managed venues with types, pricing, and operational status. The attraction create page collects venue information including title, description, attraction type selection, opening hours by day, average visit duration, suitable audiences, visit types offered, pricing, capacity limits, location with coordinates, feature tags, tips for visitors, and image gallery. The attraction edit page allows information updates and schedule modifications. The booking pages handle ticket reservations with visitor counts, visit dates, and confirmation workflow. The review pages monitor visitor experiences and enable response to feedback.

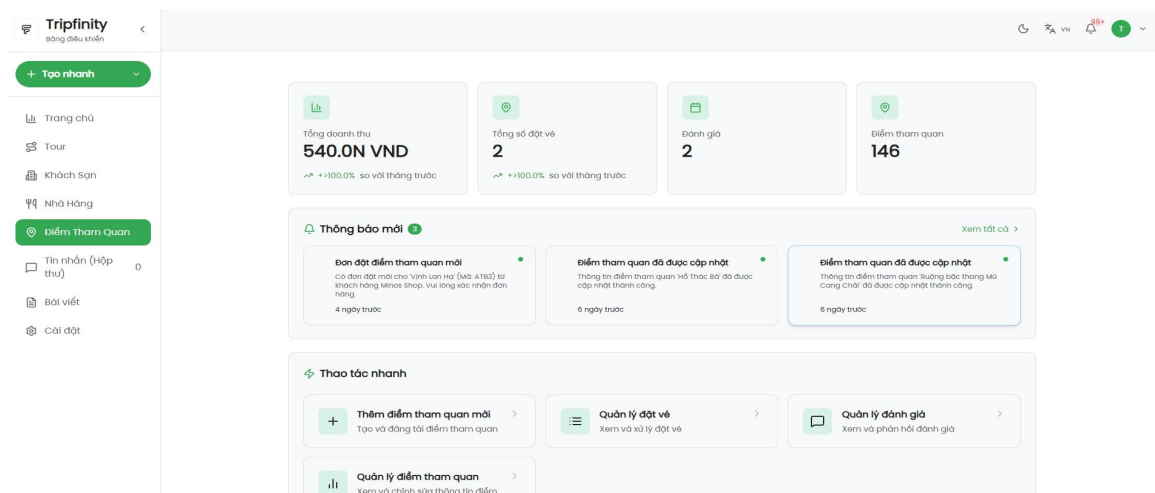


Figure 3.17. Attraction Management

3.2.6. Restaurant Management

The restaurant dashboard page presents dining business metrics including table bookings, cover counts, revenue, and customer ratings. The restaurant list page shows all establishments with cuisine types, price levels, and status. The restaurant create page provides fields for restaurant details including name, description, cuisine types (multi-select), dietary accommodations, price level, operating hours, address with map, capacity, phone, website, service options (dine-in, takeaway, delivery), payment methods, ambiance tags, menu highlights, and policy information. The restaurant edit page enables menu and detail updates.

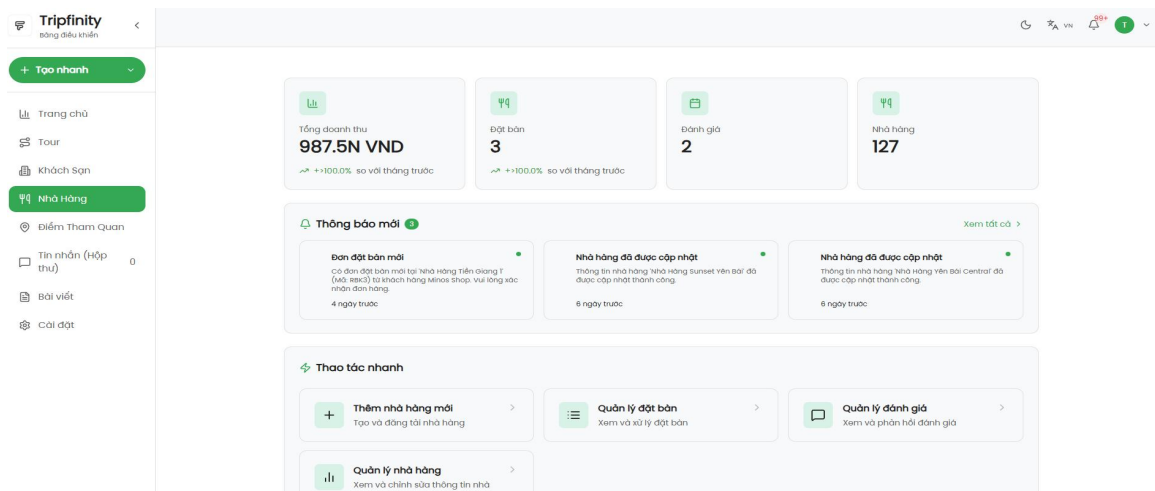


Figure 3.18. Restaurant Management

3.2.7. Blog Management

The blog list page displays all published and draft articles with view counts, publication dates, and status indicators. The create blog page provides a rich text editor for composing travel articles with title, content formatting, cover image upload, tag selection, and SEO metadata.

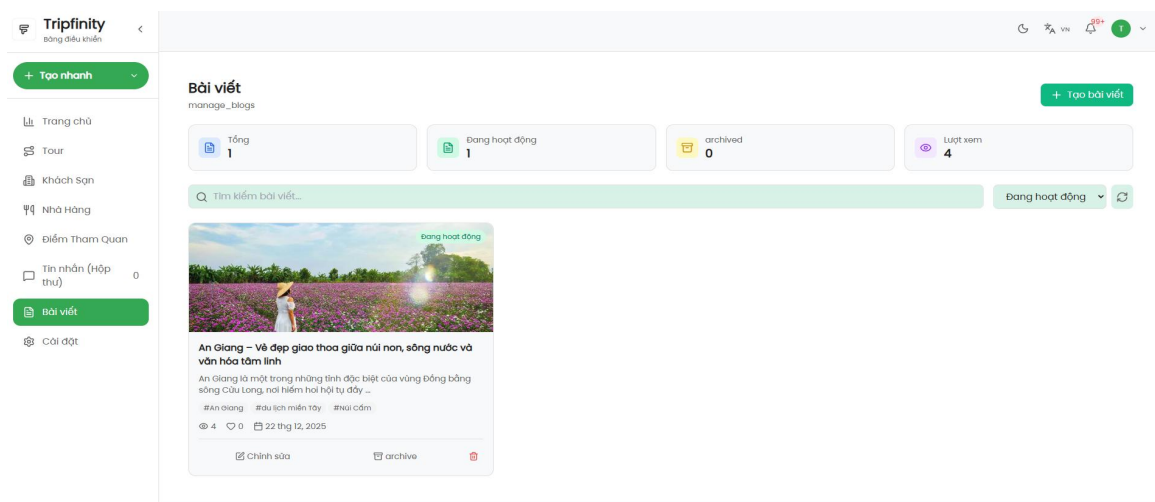


Figure 3.19. Blog Management

3.2.8. Messaging

The messages page provides a real-time chat interface for customer communication. The conversation list shows all active chats with customer names, last message previews, timestamps, and unread message counts. The chat window supports text messaging, image sharing, and file attachments. Message status indicators show sent, delivered, and read states. Quick response templates help suppliers reply efficiently to common inquiries. Notification badges alert suppliers to new incoming messages requiring attention.

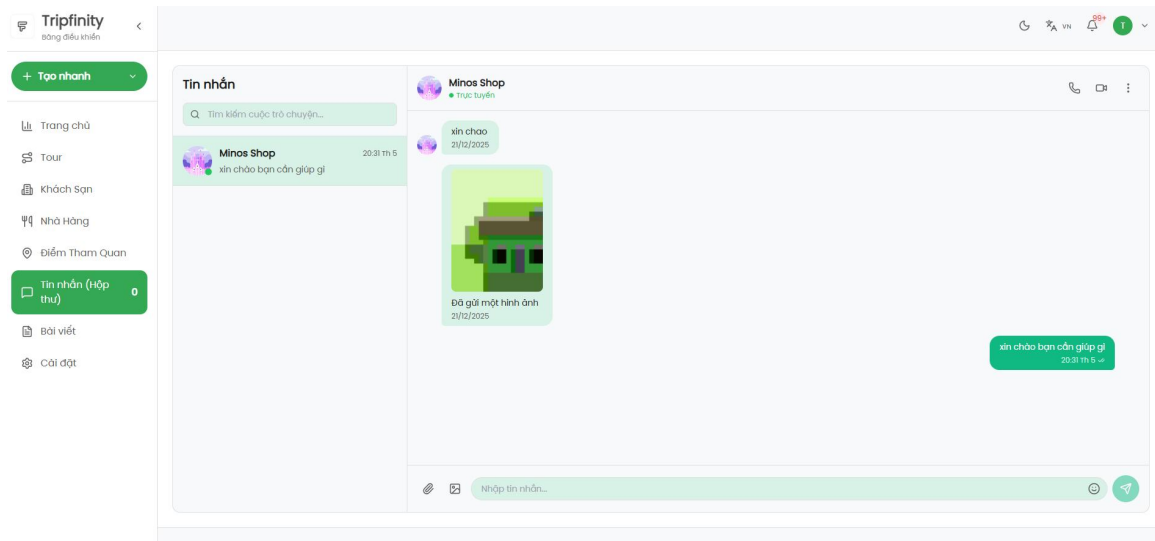


Figure 3.20. Messaging

CONCLUSIONS AND FUTURE WORK

1. ACHIEVEMENTS

Developed the Tripfinity ecosystem including a customer mobile app (Flutter), supplier web portal (React), backend (Java Spring Boot), and AI services (Python).

AI Recommendation System (Two-Tower Neural Network):

Provides personalized travel service recommendations based on user behavior.

Achieved practical accuracy of ~70-75% (test set), AUC ~80-90%.

Good overfitting control and generalization on new data.

Real-time API deployment for fast user responses.

AI Chatbot (RAG System):

Delivers natural travel consultation in Vietnamese, supporting multi-turn conversation, semantic mapping, and price intelligence.

Implemented features: context memory, Vietnamese text processing (with/without diacritics), fallback strategies when no results are found.

No automated evaluation system yet; accuracy mainly assessed via user feedback.

AI integration in App/Web: Recommendation widgets, chat interface, user interaction tracking, A/B testing, and automated retraining pipeline.

Comprehensive travel service management: Tours, hotels, attractions, restaurants, reviews, feedback, points, badges, blog, chat, notifications, and payment.

Standardized database: Stores user behavior, service information, feedback, points, badges, and interactions for AI/ML.

2. LIMITATIONS

Not yet production-scale: Some AI features are still prototypes and not optimized for high traffic.

No automated evaluation system for chatbot response quality.

Incomplete integration of real-world data sources (travel APIs, booking, etc.).

No logging, monitoring, or alerting system for AI servers.

Some advanced features (recommendation explainability, deep personalization) are unfinished.

Recommendation and chatbot accuracy still depend on data quality and real user feedback.

3. FUTURE WORK

Optimize AI models: Improve accuracy, reduce latency, experiment with new architectures (transformer-based recommender, hybrid models).

Production deployment: Migrate AI services to the cloud, integrate CI/CD, auto-scaling, and monitoring.

Expand data sources: Connect to real APIs, collect multi-dimensional data (behavior, feedback, booking).

Develop automated evaluation systems for chatbot and recommendation quality.

Upgrade UI/UX: Enhance user experience and add deeper personalization features.

Explore new directions: Explainable AI, privacy-preserving recommendation, multi-modal search (image, text, voice).

Increase real-world testing and feedback collection to improve accuracy and service quality.

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